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To cite this article: Fatimah M. Alghamdi , Ahmed M. Gemeay , Gamal A. Abd-Elmougod , Eslam Hussam & Ali T. Hammad (22 May 2026): A new regularized Poisson regression estimator for effectively modelling multicollinear count data, Statistics, DOI: [10.1080/02331888.2026.2676679](https://doi.org/10.1080/02331888.2026.2676679)

To link to this article: <https://doi.org/10.1080/02331888.2026.2676679>



Published online: 22 May 2026.



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A new regularized Poisson regression estimator for effectively modelling multicollinear count data

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ABSTRACT

Poisson regression is widely used to model count data across various fields, including healthcare, economics, engineering, and environmental research. It is designed to describe the relationship between a numerical response variable and one or more explanatory variables. A maximum likelihood estimator is typically used to estimate the parameters of a Poisson regression model. This method is highly effective under normal conditions, but when the explanatory variables are highly correlated, it violates the model's assumption of independence. This problem is known as multicollinearity, which leads to the failure to identify individual effects, variance inflation, and an increased mean squared error, thus weakening the model's reliability. Some biased estimators are introduced as an alternative to the maximum likelihood estimator. In this study, we propose a novel, alternative, and effective estimator for estimating parameters in a Poisson regression model with multicollinearity. The statistical properties of the proposed estimator are presented, and to confirm its effectiveness, theoretical comparisons with existing estimators are used, along with an extended Monte Carlo simulation under various conditions. The results validate the superiority and effectiveness of the proposed estimator. To support these results, two real-world data applications are used, which confirm the findings of both the simulation and the theoretical comparisons, demonstrating the effectiveness and reliability of the proposed estimator in estimating the parameters of a Poisson regression model under multicollinear conditions.

ARTICLE HISTORY

Received 20 February 2026

Accepted 12 May 2026

KEYWORDS

Poisson regression model; multivariate count data; multicollinearity; ridge-type estimator; Monte Carlo simulation

2020 MATHEMATICS SUBJECT CLASSIFICATIONS

62J12; 62J07; 62H12; 62J10

1. Introduction

Count regression models are used when the response variable is represented as a numerical variable or a non-negative integer. An example of such a model would be the quantity of

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defects, errors, or bugs present in a manufactured product. In software, examples include the number of road accidents, the number of machine breakdowns per month, the frequency of viral outbreaks, the number of particles or other pollutants in the air, and so on. One of the most popular and widely used models is the Poisson regression model (PRM) [1,2]. The advantage of the PRM is that it belongs to the family of generalized linear models (GLMs), describing the linear logarithmic relationship between predictive variables and event frequency while acknowledging the inherent randomness in statistical generation processes. This simplifies research for scientists studying both rare and frequent events, such as crime patterns, insurance claims, and epidemiological trends. The model provides a robust and accurate framework for practical application, offering reliable statistical inference [3,4].

The maximum likelihood estimator (MLE) is a common method for estimating unknown parameters in PRM. It is the most suitable estimator for this model, and as long as the dataset does not have excessively large or small variances, it is the standard method for solving this type of counting problem. However, in the case of multicollinearity, the mean squared error (MSE) of the MLE becomes unstable due to the high variance of the regression coefficients, and inferences based on the MLE may not be reliable. Another effect of multicollinearity is that it widens confidence intervals and reduces statistical power, increasing the likelihood of Type II errors in hypothesis testing for the parameters. Furthermore, the uncertainty in the estimated coefficients increases as a result of the increased variance of the coefficients caused by multicollinearity [5–7]. To address the problem of multicollinearity in regression models, several researchers have introduced biased estimators based on contraction coefficients to balance bias and variance, thereby reducing variance inflation and significantly lowering the mean squared error [8–10]. Hoerl and Kennard [11] were pioneers in this approach, and subsequently, many researchers, including Dawoud and Kibria [12], Idowu [13], Lukman et al. [14,15], Liu [16,17], Dawoud and Eledum [18], Abonazel et al. [19], and Dawoud et al. [20], have further developed these estimators.

Hoerl and Kennard [11] proposed an ancient but effective method for mitigating multicollinearity by adding a shrinkage parameter ($k > 0$) for the diagonal elements of the design matrix. To estimate the biasing parameter k for the ridge regression estimator, Several efforts have been made by researchers, including Kibria [21], Khalaf and Shukur [22], and Lukman and Ayinde [23]. Following these previous works, Månsson and Shukur [24] proposed the ridge regression estimator for PRM based on several shrinkage indicators discussed in earlier studies. Alternatively, Liu [16] proposed another estimator with a linear shrinkage parameter d , and building upon this work, Månsson et al. [25] introduced the Liu estimator for PRM and suggested some suitable values for the Liu estimator within the context of the model. The ridge regression estimator and the Liu estimator are both examples of single-parameter estimators. Recently, Lukman et al. [14] developed the modified ridge-type estimator with two parameters: k and d for the linear regression model. Following this work, Lukman et al. [26] defined the modified ridge-type estimator for PRM, achieving better results and better handling of multicollinearity.

Several recent studies have proposed new biased estimators based on two shrinkage parameters for linear and generalized regression models to address the problem of multicollinearity. These studies have shown that estimators using two shrinkage parameters yield better results than those using only one shrinkage parameter. Among these studies

are Alqasem et al. [27], Hammad et al. [28], and Alghamdi et al. [29]. This article aims to introduce a novel version of a ridge-type estimator for the PRM. It presents the statistical characteristics of the proposed estimator and compares them with existing estimators, selecting shrinkage parameters. The effectiveness of the proposed estimator is verified using both simulations and real-world applications.

The remainder of this article is divided as follows: Section 2 reviews the PRM and the MLE to estimate parameters in the model, along with considering current biased estimators. Section 3 discusses the proposed estimator and its characteristics, along with theoretical comparisons and parameter selection methods for estimators. Section 4 presents a Monte Carlo simulation. Section 5 provides real-world examples of the proposed estimator using real data. Finally, Section 6 presents conclusions and future directions.

2. Methodology

This section presents a literature review of the PRM and its traditional estimation method, the MLE method, and current biased estimation methods for addressing multicollinearity in the model.

2.1. Poisson regression model

PRM is the most common count regression model, analyzing response variables consisting of non-negative integer data. Assume the response variable y_i follows a Poisson distribution with a parameter of μ_i and is denoted as $y_i \sim \text{Poisson}(\mu_i)$. Then its probability mass function (PMF) is given by:

$$f(y_i) = \frac{e^{-\mu_i} \mu_i^{y_i}}{y_i!}; \quad y_i = 0, 1, 2, \dots; \quad i = 1, 2, \dots, n. \quad (1)$$

The PRM relates the parameter to a linear predictor using the canonical log-link function, which is typically formulated as follows:

$$\mu_i = \exp(x_i' \beta),$$

where x_i represents the i th row of an $n \times p$ matrix X containing p non-stochastic explanatory variables, and β is a $p \times 1$ vector of unknown parameters. Given this parameterization, the log-likelihood function for the model can be expressed as follows:

$$l(\mu; y) = \sum_{i=1}^n \{y_i \ln(\mu_i) - \mu_i - \ln(y_i!)\}, \quad (2)$$

or equivalently,

$$l(\beta; y) = \sum_{i=1}^n \{y_i(x_i' \beta) - \exp(x_i' \beta) - \ln(y_i!)\}. \quad (3)$$

The parameters of the PRM are estimated using MLE. Due to the non-linearity of MLE, it is commonly achieved using an iteratively reweighted least squares algorithm, which gives

the Poisson maximum likelihood estimator (PMLE) in the last step of the algorithm as follows:

$$\hat{\beta}^{\text{PMLE}} = (X' \hat{U} X)^{-1} X' \hat{U} \hat{c}, \quad (4)$$

where $\hat{U} = \text{diag}(\hat{\mu}_1, \hat{\mu}_2, \dots, \hat{\mu}_n)$ is a diagonal weight matrix, and $\hat{c}$ is an adjusted response vector with elements $c_i = \log(\hat{\mu}_i) + (y_i - \hat{\mu}_i)/\hat{\mu}_i$. Both $\hat{U}$ and c are updated iteratively using a procedure such as Fisher's scoring.

To calculate the statistical properties of $\hat{\beta}^{\text{PMLE}}$, we consider the spectral decomposition of the information matrix, $X' \hat{U} X = \theta Q \theta'$, where θ is an orthogonal matrix of eigenvectors of $X' \hat{U} X$ and $Q = \text{diag}(q_1, q_2, \dots, q_p)$ contains the corresponding eigenvalues $q_1 \geq q_2 \geq \dots \geq q_p > 0$. Let $\alpha = \theta' \hat{\beta}^{\text{PMLE}}$.

The variance-covariance matrix, the matrix mean squared error (MMSE), and the MSE of the PMLE are, respectively, given by:

$$\text{Cov}(\hat{\beta}^{\text{PMLE}}) = (X' \hat{U} X)^{-1}, \quad (5)$$

$$\text{MMSE}(\hat{\beta}^{\text{PMLE}}) = \theta Q^{-1} \theta', \quad (6)$$

$$\text{MSE}(\hat{\beta}^{\text{PMLE}}) = \sum_{j=1}^p \frac{1}{q_j}. \quad (7)$$

In the case of highly correlated explanatory variables, this issue is known as multicollinearity, making the matrix $X' \hat{U} X$ become ill-conditioned, which leads to one or more eigenvalues q_j being tiny. According to Equation (7), such behaviour inflates the MSE and thus the variance of the PMLE, undermining the stability of the estimates. It is this problem of multicollinearity-induced variance inflation that necessitates the development of robust estimation techniques.

2.2. Poisson ridge regression estimator

In order to address the multicollinearity problem in PRM, Mansson and Shukur [24] proposed the Poisson ridge regression estimator (PRRE), which is a modification of the PMLE that adds a small value (k) to the diagonal elements of the design matrix. The PRRE is defined to PRM as follows:

$$\hat{\beta}^{\text{PRRE}} = (X' \hat{U} X + kI)^{-1} X' \hat{U} X \hat{\beta}^{\text{PMLE}} \quad k > 0, \quad (8)$$

where k is the ridge parameter and I represent the identity matrix of order $p \times p$. When k equals zero in Equation (8), the PRRE is equivalent to the PMLE. The bias vector and covariance matrix of PRRE in Equation (8) are defined as follows:

$$\text{Bias}(\hat{\beta}^{\text{PRRE}}) = -k \theta Q_k^{-1} \alpha, \quad (9)$$

$$\text{Cov}(\hat{\beta}^{\text{PRRE}}) = \theta Q_k^{-1} Q Q_k^{-1} \theta'. \quad (10)$$

The MSE and MMSE formulations related to the PRRE are as follows:

$$\text{MMSE}(\hat{\beta}^{\text{PRRE}}) = \theta Q_k^{-1} Q Q_k^{-1} \theta' + k^2 \theta Q_k^{-1} \alpha \alpha' Q_k^{-1} \theta', \quad (11)$$

$$\text{MSE}(\hat{\beta}^{\text{PRRE}}) = \sum_{j=1}^p \frac{q_j}{(q_j + k)^2} + k^2 \sum_{j=1}^p \frac{\alpha_j^2}{(q_j + k)^2}, \quad (12)$$

where the matrix $Q_k = \text{diag}(q_1 + k, q_2 + k, \dots, q_p + k)$ and symbol α represents the j th element of the vector $\alpha = (\alpha_1, \alpha_2, \dots, \alpha_p)'$. This vector is derived by multiplying the transpose of the matrix θ with the parameter vector β .

2.3. Poisson Liu regression estimator

For more extended biased estimators in the PRM, Månsson et al. [25] proposed an alternative estimator to PMLE, called the Poisson Liu regression estimator (PLRE). This estimator addresses multicollinearity more effectively than the PMLE. The PLRE is defined with the PRM as follows:

$$\hat{\beta}^{\text{PLRE}} = (X' \hat{U} X + I)^{-1} (X' \hat{U} X + dI) \hat{\beta}^{\text{PMLE}} \quad 0 < d < 1, \quad (13)$$

where the d is the Liu shrinkage estimator. The behaviour of the PLRE depends on the value of d , where if $d = 1$, then the PLRE is equivalent to the PMLE. The bias vector and covariance matrix of PLRE in Equation (13) are defined as follows:

$$\text{Bias}(\hat{\beta}^{\text{PLRE}}) = (d - 1) \theta Q_I^{-1} \alpha, \quad (14)$$

$$\text{Cov}(\hat{\beta}^{\text{PLRE}}) = \theta Q_I^{-1} Q_d Q^{-1} Q_d Q_I^{-1} \theta'. \quad (15)$$

The MSE and MMSE formulations related to the PLRE are as follows:

$$\text{MMSE}(\hat{\beta}^{\text{PLRE}}) = \theta Q_I^{-1} Q_d Q^{-1} Q_d Q_I^{-1} \theta' + (d - 1)^2 \theta Q_I^{-1} \alpha \alpha' Q_I^{-1} \theta', \quad (16)$$

$$\text{MSE}(\hat{\beta}^{\text{PLRE}}) = \sum_{j=1}^p \frac{(q_j + d)^2}{q_j (q_j + 1)^2} + (d - 1)^2 \sum_{j=1}^p \frac{\alpha_j^2}{(q_j + 1)^2}, \quad (17)$$

where the matrix $Q_I = \text{diag}(q_1 + 1, q_2 + 1, \dots, q_p + 1)$ and the matrix $Q_d = \text{diag}(q_1 + d, q_2 + d, \dots, q_p + d)$.

2.4. Poisson modified ridge-type estimator

Recently, Lukman et al. [26] proposed a modified version of the ridge regression estimator with two regularization parameters, which achieved more stable coefficient estimates in the PRM. This estimator, named the Poisson modified ridge-type estimator (PMRTE) and has the following formula:

$$\hat{\beta}^{\text{PMRTE}} = (X' \hat{U} X + k(1 + d)I)^{-1} X' \hat{U} X \hat{\beta}^{\text{PMLE}} \quad 0 < d < 1, k > 0, \quad (18)$$

where d and k represent the modified ridge-type shrinkage parameters. It is important to note that if $d = 0$, then the PMRTE is equivalent to the PRRE and if $k = 0$, then the

PMRTE is equivalent to the PMLE. The bias vector and covariance matrix of PMRTE in Equation (18) are defined as follows:

$$\text{Bias}(\hat{\beta}^{\text{PMRTE}}) = -k(d+1)\theta Q_{kd}^{-1}\alpha, \quad (19)$$

$$\text{Cov}(\hat{\beta}^{\text{PMRTE}}) = \theta Q_{kd}^{-1} Q Q_{kd}^{-1} \theta'. \quad (20)$$

The MSE and MMSE formulations related to the PMRTE are as follows:

$$\text{MMSE}(\hat{\beta}^{\text{PMRTE}}) = \theta Q_{kd}^{-1} Q Q_{kd}^{-1} \theta' + k^2(d+1)^2 \theta Q_{kd}^{-1} \alpha \alpha' Q_{kd}^{-1} \theta', \quad (21)$$

$$\text{MSE}(\hat{\beta}^{\text{PMRTE}}) = \sum_{j=1}^p \frac{q_j}{(q_j + k(d+1))^2} + k^2(d+1)^2 \sum_{j=1}^p \frac{\alpha_j^2}{(q_j + k(d+1))^2}, \quad (22)$$

where $Q_{kd} = \text{diag}(q_1 + k(1+d), q_2 + k(1+d), \dots, q_p + k(1+d))$.

3. Proposed estimator

In this section, we present a new version of the ridge-type estimator: a modified version of the ridge estimator with the addition of an extra shrinkage parameter (d) alongside (k). We present the statistical properties of the proposed estimator, theoretically compare it with existing estimators, and discuss parameter selection methods for estimators.

3.1. Poisson adjusted ridge-type estimator

Based on the work of Hoerl and Kennard [11], Lukman et al. [14] proposed a new version of the ridge estimator, which they called the modified ridge estimator. Later, Lukman et al. [26] proposed the modified ridge estimator within the PRM model in the presence of multicollinearity. Therefore, in this paper, we present a novel version of the ridge-type estimator; this version is a modification of the ridge regression estimator by adding another shrinkage parameter d alongside k . This leads to reduced variance and a better bias-variance trade-off compared to existing estimators. The proposed adjusted ridge-type estimator introduces a new formulation incorporating the constraint $(kd/k^{1/2}(1+d)^{1/2}) \hat{\beta} = k^{1/2}(1+d)^{1/2} \beta + \varepsilon'$. This expansion strategy results in a biased but more stable estimator, as it offers a better balance between bias and variance and is particularly effective in high multicollinearity scenarios.

This new estimator is defined for the PRM, named the new Poisson adjusted ridge-type estimator (PARTE), which is defined as follows:

$$\hat{\beta}^{\text{PARTE}} = (X' \hat{U} X + k(1+d)I)^{-1} (X' \hat{U} X + kdI) \hat{\beta}^{\text{PMLE}}, \quad 0 < d < 1, k > 0, \quad (23)$$

where k and d are the adjusted ridge-type shrinkage parameters. The proposed estimator is a generalization of some existing estimators, as they are special cases of it. When $k = 0$, then the PARTE reduced to the PMLE and when $d = 0$, then the PARTE reduced to the PRRE. The bias vector and variance-covariance matrix of the proposed estimator are

given by

$$\begin{aligned} \text{Bias}(\hat{\beta}^{\text{PARTE}}) &= E(\hat{\beta}^{\text{PARTE}}) - \beta = \left((X' \hat{U}X + k(1+d)I)^{-1} (X' \hat{U}X + kd) \right) \beta - \beta \\ &= -k\theta Q_{kd}^{-1} \alpha, \end{aligned} \quad (24)$$

$$\begin{aligned} \text{Cov}(\hat{\beta}^{\text{PARTE}}) &= E \left(\left[\hat{\beta}^{\text{PARTE}} - E(\hat{\beta}^{\text{PARTE}}) \right] \left[\hat{\beta}^{\text{PARTE}} - E(\hat{\beta}^{\text{PARTE}}) \right]' \right) \\ &= \left[(X' \hat{U}X + k(1+d)I)^{-1} (X' \hat{U}X + kd) \right] \text{Cov}(\hat{\beta}^{\text{PMLE}}) \\ &\quad \times \left[(X' \hat{U}X + k(1+d)I)^{-1} (X' \hat{U}X + kd) \right]' \\ &= \theta Q_{kd}^{-1} Q_N Q Q_{kd}^{-1} Q_N \theta'. \end{aligned} \quad (25)$$

Using Equations (24) and (25) to calculate MMSE and MSE of the PARTE as follows:

$$\begin{aligned} \text{MMSE}(\hat{\beta}^{\text{PARTE}}) &= \text{Cov}(\hat{\beta}^{\text{PARTE}}) + \text{Bias}(\hat{\beta}^{\text{PARTE}}) \text{Bias}(\hat{\beta}^{\text{PARTE}})' \\ &= \theta Q_{kd}^{-1} Q_N Q Q_{kd}^{-1} Q_N \theta' + k^2 \theta Q_{kd}^{-1} \alpha \alpha' Q_{kd}^{-1} \theta', \end{aligned} \quad (26)$$

$$\begin{aligned} \text{MSE}(\hat{\beta}^{\text{PARTE}}) &= E \left[(\hat{\beta}^{\text{PARTE}} - \beta)' (\hat{\beta}^{\text{PARTE}} - \beta) \right] = \text{tr} \left(\text{MMSE}(\hat{\beta}^{\text{PARTE}}) \right) \\ &= \sum_{j=1}^p \frac{(q_j + kd)^2}{q_j(q_j + k(1+d))^2} + k^2 \sum_{j=1}^p \frac{\alpha_j^2}{(q_j + k(1+d))^2}, \end{aligned} \quad (27)$$

where $\text{tr}(\cdot)$ is the trace of the matrix and $Q_N = \text{diag}(q_1 + kd, q_2 + kd, \dots, q_p + kd)$.

3.2. Comparison of the estimators

Lemma 3.1: Let M be a positive definite (p.d.) matrix, γ be a vector of nonzero constants and c be a positive constant. Then $cM - \gamma \gamma' > 0$ if and only if $\gamma' M^{-1} \gamma < c$ [30].

Theorem 3.1: Within the framework of the PRM, Let b_1 is the bias of the PARTE, $k > 0$, and $0 < d < 1$. Then $\text{MMSE}(\hat{\beta}^{\text{PMLE}}) - \text{MMSE}(\hat{\beta}^{\text{PARTE}}) > 0$ if and only if (iff)

$$b_1' [\theta Q^{-1} \theta' - \theta Q_{kd}^{-1} Q_N Q Q_{kd}^{-1} Q_N \theta'] b_1 < 1.$$

Proof: The difference between the MMSE functions of the PMLE and PARTE estimators is given by

$$\text{MMSE}(\hat{\beta}^{\text{PMLE}}) - \text{MMSE}(\hat{\beta}^{\text{PARTE}}) = \theta Q^{-1} \theta' - \theta Q_{kd}^{-1} Q_N Q Q_{kd}^{-1} Q_N \theta' - b_1 b_1'. \quad (28)$$

The above expression can be described in terms of MSE, which can be written as follows:

$$\begin{aligned} \Delta_1 &= \text{MSE}(\hat{\beta}^{\text{PMLE}}) - \text{MSE}(\hat{\beta}^{\text{PARTE}}) \\ &= \theta \text{diag} \left\{ \frac{1}{q_j} - \frac{(q_j + kd)^2}{q_j(q_j + k(1+d))^2} \right\}_{j=1}^r \theta' - b_1 b_1'. \end{aligned}$$

After simplification, we obtain:

$$\Delta_1 = \theta \operatorname{diag} \left\{ \frac{(q_j + k(1+d))^2 - (q_j + kd)^2}{q_j(q_j + k(1+d))^2} \right\}_{j=1}^r \theta' - b_1 b'_1.$$

The expression written in Equation (28) i.e., $\theta [Q^{-1} - Q_{kd}^{-1} Q_N Q Q_{kd}^{-1} Q_N] \theta'$ is p.d. iff $(q_j + k(1+d))^2 - (q_j + kd)^2 > 0$ by simplifying this inequality, one gets $2kq_j + k^2(1+d) > 0$. Thus if $k > 0$, $0 < d < 1$, then the theorem is completed by Lemma 3.1. ■

Theorem 3.2: *Within the framework of the PRM, Let b_1 is the bias of the PARTE, b_2 is the bias of the PRRE, $k > 0$, and $0 < d < 1$. Then $MMSE(\hat{\beta}^{PRRE}) - MMSE(\hat{\beta}^{PARTE}) > 0$ iff*

$$b'_1 [\theta Q_k^{-1} Q Q_k^{-1} \theta' - \theta Q_{kd}^{-1} Q_N Q Q_{kd}^{-1} Q_N \theta' + b_2 b'_2] b_1 < 1.$$

Proof: The difference between the MMSE functions of the PRRE and PARTE estimators is given by

$$\begin{aligned} MMSE(\hat{\beta}^{PRRE}) - MMSE(\hat{\beta}^{PARTE}) &= \theta Q_k^{-1} Q Q_k^{-1} \theta' - \theta Q_{kd}^{-1} Q_N Q Q_{kd}^{-1} Q_N \theta' \\ &\quad + b_2 b'_2 - b_1 b'_1. \end{aligned} \quad (29)$$

The above expression can be described in terms of MSE, which can be written as follows:

$$\begin{aligned} \Delta_2 &= MSE(\hat{\beta}^{PRRE}) - MSE(\hat{\beta}^{PARTE}) \\ &= \theta \operatorname{diag} \left\{ \frac{q_j + k^2 \alpha_j^2}{(q_j + k)^2} - \frac{(q_j + kd)^2}{q_j(q_j + k(1+d))^2} \right\}_{j=1}^r \theta' - b_1 b'_1. \end{aligned}$$

After simplification, we obtain:

$$\begin{aligned} \Delta_2 &= \theta \operatorname{diag} \left\{ \frac{q_j^2(q_j + k(1+d))^2 - (q_j + k)^2(q_j + kd)^2 + k^2 \alpha_j^2 q_j(q_j + k(1+d))^2}{q_j(q_j + k)^2(q_j + k(1+d))^2} \right\}_{j=1}^r \\ &\quad \times \theta' - b_1 b'_1. \end{aligned}$$

The expression written in Equation (29) i.e., $\theta [Q_k^{-1} Q Q_k^{-1} - Q_{kd}^{-1} Q_N Q Q_{kd}^{-1} Q_N] \theta'$ is p.d. iff $q_j^2(q_j + k(1+d))^2 - (q_j + k)^2(q_j + kd)^2 + k^2 \alpha_j^2 q_j(q_j + k(1+d))^2 > 0$ by simplifying this inequality, one gets $k^2 q_j^2(\alpha_j^2 q_j - 2d) + 2k^3(1+d)q_j(\alpha_j^2 q_j - d) + k^4(\alpha_j^2(1+d)^2 q_j - d^2) > 0$. Thus if $k > 0$, $0 < d < 1$, then the theorem is completed by Lemma 3.1. ■

Theorem 3.3: *Within the framework of the PRM, Let b_1 is the bias of the PARTE, b_3 is the bias of the PLRE, $k > 0$, and $0 < d < 1$. Then $MMSE(\hat{\beta}^{PLRE}) - MMSE(\hat{\beta}^{PARTE}) > 0$ iff*

$$b'_1 [\theta Q_I^{-1} Q_d Q^{-1} Q_d Q_I^{-1} \theta' - \theta Q_{kd}^{-1} Q_N Q Q_{kd}^{-1} Q_N \theta' + b_3 b'_3] b_1 < 1.$$

Proof: The difference between the MMSE functions of the PLRE and PARTE estimators is given by

$$\begin{aligned} MMSE\left(\hat{\beta}^{PLRE}\right) - MMSE\left(\hat{\beta}^{PARTE}\right) &= \theta Q_I^{-1} Q_d Q^{-1} Q_d Q_I^{-1} \theta' - \theta Q_{kd}^{-1} Q_N Q Q_{kd}^{-1} Q_N \theta' \\ &\quad + b_3 b'_3 - b_1 b'_1. \end{aligned} \quad (30)$$

The above expression can be described in terms of MSE, which can be written as follows:

$$\begin{aligned} \Delta_3 &= MSE\left(\hat{\beta}^{PLRE}\right) - MSE\left(\hat{\beta}^{PARTE}\right) \\ &= \theta \text{diag} \left\{ \frac{(q_j + d)^2 + q_j(d-1)^2 \alpha_j^2}{q_j(q_j + 1)^2} - \frac{(q_j + kd)^2}{q_j(q_j + k(1+d))^2} \right\}_{j=1}^r \theta' - b_1 b'_1. \end{aligned}$$

After simplification, we obtain:

$$\begin{aligned} \Delta_3 &= \theta \text{diag} \left\{ \frac{\left[(q_j + d)^2 + q_j(d-1)^2 \alpha_j^2 \right] (q_j + k(1+d))^2 - (q_j + 1)^2 (q_j + kd)^2}{q_j(q_j + 1)^2 (q_j + k(1+d))^2} \right\}_{j=1}^r \\ &\quad \times \theta' - b_1 b'_1. \end{aligned}$$

The expression written in Equation (30) i.e., $\theta [Q_I^{-1} Q_d Q^{-1} Q_d Q_I^{-1} - Q_{kd}^{-1} Q_N Q Q_{kd}^{-1} Q_N] \theta'$ is p.d. iff $\left[(q_j + d)^2 + q_j(d-1)^2 \alpha_j^2 \right] (q_j + k(1+d))^2 - (q_j + 1)^2 (q_j + kd)^2 > 0$ by simplifying this inequality, one gets $q_j(d-1)^2 \alpha_j^2 (q_j + k(1+d))^2 + [q_j(k+d-1) + d^2 k] [2q_j^2 + (k+d+1+2kd)q_j + dk(2+d)] > 0$. Thus if $k > 0$, $0 < d < 1$, then the theorem is completed by Lemma 3.1. ■

Theorem 3.4: Within the framework of the PRM, Let b_1 is the bias of the PARTE, b_4 is the bias of the PMRTE, $k > 0$, and $0 < d < 1$. Then $MMSE\left(\hat{\beta}^{PMRTE}\right) - MMSE\left(\hat{\beta}^{PARTE}\right) > 0$ iff

$$b'_1 [\theta Q_{kd}^{-1} Q Q_{kd}^{-1} \theta' - \theta Q_{kd}^{-1} Q_N Q Q_{kd}^{-1} Q_N \theta' + b_4 b'_4] b_1 < 1.$$

Proof: The difference between the MMSE functions of the PMRTE and PARTE estimators is given by

$$\begin{aligned} MMSE\left(\hat{\beta}^{PMRTE}\right) - MMSE\left(\hat{\beta}^{PARTE}\right) &= \theta Q_{kd}^{-1} Q Q_{kd}^{-1} \theta' - \theta Q_{kd}^{-1} Q_N Q Q_{kd}^{-1} Q_N \theta' \\ &\quad + b_4 b'_4 - b_1 b'_1. \end{aligned} \quad (31)$$

The above expression can be described in terms of MSE, which can be written as follows:

$$\begin{aligned} \Delta_4 &= MSE\left(\hat{\beta}^{PMRTE}\right) - MSE\left(\hat{\beta}^{PARTE}\right) \\ &= \theta \text{diag} \left\{ \frac{q_j + k^2(d+1)^2 \alpha_j^2}{(q_j + k(d+1))^2} - \frac{(q_j + kd)^2}{q_j(q_j + k(1+d))^2} \right\}_{j=1}^r \theta' - b_1 b'_1. \end{aligned}$$

After simplification, we obtain:

$$\Delta_4 = \theta \operatorname{diag} \left\{ \frac{q_j(q_j + k^2(d+1)^2\alpha_j^2) - (q_j + kd)^2}{q_j(q_j + k(1+d))^2} \right\}_{j=1}^r \theta' - b_1 b_1'.$$

The expression written in Equation (31) i.e., $\theta [Q_{kd}^{-1} Q Q_{kd}^{-1} - Q_{kd}^{-1} Q_N Q Q_{kd}^{-1} Q_N] \theta'$ is p.d. iff $q_j(q_j + k^2(d+1)^2\alpha_j^2) - (q_j + kd)^2 > 0$ by simplifying this inequality, one gets $k[q_j k(d+1)^2\alpha_j^2 - 2q_j d - kd^2] > 0$. Thus if $k > 0$, $0 < d < 1$, then the theorem is completed by Lemma 3.1. ■

3.3. Selection of biasing parameter

Biased estimators rely primarily on shrinkage parameters, which significantly impact estimator performance as they balance bias and variance to minimize the MSE value. Therefore, finding optimal parameter values for biased estimators is critical.

Following the works of Abdelwahab et al. [8], Lukman et al. [26], Amin et al. [31], and Aladeitan et al. [32], the best values of k for PRRE are determined as follows:

$$\hat{k}_1 = \frac{1}{\sum_{j=1}^p \hat{\alpha}_j^2}, \quad (32)$$

$$\hat{k}_1 = \min_j \left(\frac{1}{\hat{\alpha}_j^2} \right). \quad (33)$$

Building upon the works of Qasim et al. [33], Lukman et al. [26], and Abdelwahab et al. [8], the best values of d for PLRE are determined as follows:

$$\hat{d}_1 = \min_j \left(\frac{\hat{\alpha}_j^2 - 1}{\frac{1}{q_j} + \hat{\alpha}_j^2} \right), \quad (34)$$

$$\hat{d}_2 = \max_j \left(\frac{\hat{\alpha}_j^2 - 1}{\frac{1}{q_j} + \hat{\alpha}_j^2} \right). \quad (35)$$

According to the works of Lukman et al. [14,26], Alghamdi et al. [9], and Alqasem et al. [34], the best values of k and d for PMRTE are determined as follows:

$$\hat{k}_1 = \frac{1}{\max_j(\hat{\alpha}_j^2)}, \quad \hat{k}_2 = \min_j \left(\frac{1}{(1 + \hat{d})\hat{\alpha}_j^2} \right), \quad (36)$$

$$\hat{d} = \min_j \left(\frac{q_j \hat{\alpha}_j^2}{1 + q_j \hat{\alpha}_j^2} \right). \quad (37)$$

For the proposed PARTE, we find the values of k and d that minimize the MSE in Equation (27). Differentiating MSE with respect to both k and d , and setting the partial

derivatives to zero, i.e., $\frac{\partial \text{MSE}}{\partial k} = 0$ and $\frac{\partial \text{MSE}}{\partial d} = 0$. By starting with an initial value of $\hat{d} = \min_j \left(\frac{q_j \hat{\alpha}_j^2}{1 + q_j \hat{\alpha}_j^2} \right)$ yields the following optimality value for k shrinkage parameters:

$$\hat{k}_1 = \min_j \left(\frac{q_j}{\hat{\alpha}_j^2 q_j - \hat{d}} \right)^{\frac{1}{2}}, \quad (38)$$

$$\hat{k}_2 = \min_j \left(\frac{2q_j}{\hat{\alpha}_j^2 q_j - \hat{d}} \right), \quad (39)$$

$$\hat{k}_3 = \frac{1}{p} \text{mean}_j \left(\frac{1}{\hat{\alpha}_j^2 q_j - \hat{d}} \right). \quad (40)$$

The optimal value for d is given by:

$$\hat{d}_r = \min_j \left(\frac{q_j (\hat{k}_r \max_j (\hat{\alpha}_j^2) - 1)}{\hat{k}_r} \right), \quad \text{for } r = 1, 2, 3. \quad (41)$$

If $\hat{d}$ in Equation (41) falls outside the interval $[0, 1]$, using the initial value of $\hat{d} = \min_j \left(\frac{q_j \hat{\alpha}_j^2}{1 + q_j \hat{\alpha}_j^2} \right)$.

4. The Monte Carlo experiment

This section presents a condition for applying Monte Carlo simulation to compare biased estimators within a PRM under conditions of multicollinearity. This section covers data generation, discusses various factors that affect the simulation, outlines the criteria for comparing estimators, presents the simulation results, and provides commentary.

4.1. The design of experiment

The response variable y_i is generated from a Poisson distribution $P(\mu_i)$, where the mean μ_i is defined as:

$$\mu_i = \exp(\beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip}), \quad i = 1, 2, \dots, n. \quad (42)$$

The initial values of the parametric values are chosen as $\sum_{j=1}^{p-1} \beta_j^2 = 1$, which is very commonly used in simulation studies; see Qasim et al. [33] Alghamdi et al. [9], and Alqasem et al. [34]. Multicollinearity is controlled by the correlations between variables, which are denoted by ρ . Correlations between explanatory variables are generated using the following formula:

$$x_{ij} = \sqrt{(1 - \rho^2)} f_{ij} + \rho f_{i(j+1)}, \quad i = 1, 2, \dots, n, \quad j = 1, 2, \dots, p. \quad (43)$$

where f_{ij} represents the pseudo-random numbers that are generated from the standard normal distribution; moreover, five different values of multicollinearity, (ρ) 0.80, 0.85, 0.90,

0.95, and 0.99, are considered. Six values for the sample size (n), ranging from 30 to 400, were used, as well as three values for the explanatory variables (p): 3, 7, and 12. Finally, following the works of Lukman et al. [14], Abdelwahab et al. [8], and Alqasem et al. [27] three different values for β_0 were used: -1 , 0 , and 1 . These values are chosen to represent three qualitatively distinct baseline scenarios: a negative baseline effect ($\beta_0 = -1$), a null baseline effect ($\beta_0 = 0$), and a positive baseline effect ($\beta_0 = 1$), corresponding to low, moderate, and relatively higher baseline mean counts of the Poisson response, respectively. These other cases provide us with many scenarios that simulate real-world data.

The estimated MSE criterion is used to compare the proposed PARTE with existing estimators (PMLE, PRRE, PLRE, and PMRTE)

$$\text{MSE}(\hat{\beta}) = \frac{1}{R} \sum_{i=1}^R (\hat{\beta}_i - \beta)' (\hat{\beta}_i - \beta), \quad (44)$$

The number of iterations in the Monte Carlo simulation is represented by ($R = 5000$). All simulations and calculations were performed using R version 4.5.2. The large number of iterations ensures stable and reliable performance estimates.

4.2. Simulation results

Tables 1–9 show the estimated MSE values for different biased estimators in the PRMs. The estimated MSEs are also affected by factors such as degrees of multicollinearity, sample size, intercept values, and the number of explanatory variables to be considered. The primary aim is to assess the efficacy of PMLE, PRRE, PLRE, PMRTE, and PARTE in mitigating the inflation of MSE induced by degrees of multicollinearity among explanatory variables. The main observations drawn from the simulation results are summarized below:

One of the key findings is the effect of multicollinearity on the MSE value and estimation accuracy. Multicollinearity makes estimation less accurate and leads to a substantial increase in the MSE, particularly with a small sample size. However, a greater sample size decreases the MSE, which is crucial to improving estimation accuracy. In addition, as the number of explanatory variables increases, their interactions also increase, and it is hard to separate effects, which also disturbs estimation accuracy and raises the MSE. We also find that the intersection value (β_0) has an impact on the MSE; an increase in β_0 from -1 to $+1$ results in a reduction in the MSE.

The value of β_0 plays a crucial role in the simulations because it determines the baseline mean for the Poisson response variable, thereby affecting how strongly multicollinearity affects the bias-variance dilemma. For $\beta_0 = -1$ (Tables 1, 4, and 7), the expected count is extremely low, and the log-link function is highly nonlinear at the boundary, which causes numerical instability and yields a high MSE for the PMLE under strong multicollinearity ($\rho \geq 0.95$) and small sample sizes. By contrast, $\beta_0 = 1$ (Tables 3, 6, 9) increases the expected counts, making the log-likelihood surface more ‘convex’; multicollinearity plays a relatively smaller role, and the PMLE performs reasonably well; the gains from all of the biased estimators are minimal. The middle case of $\beta_0 = 0$ (Tables 2, 5, 8) is the baseline case: variance is inflated by multicollinearity, but improvements in MSE are clearly visible for all of the bias-reduction methods (particularly when PARTE uses adaptive parameters) without the extreme fluctuations seen when $\beta_0 = -1$. Therefore, β_0 is not simply a

Table 1. Estimated MSE values for biased estimators in PRM under multicollinearity when $p = 3$ and $\beta_0 = -1$.

ρ	n	PMLE			PRRE			PLRE			PMRTE			PARTE		
		-	$\hat{k}_1$	$\hat{k}_2$	$\hat{d}_1$	$\hat{d}_2$	$\hat{k}_1, \hat{d}$	$\hat{k}_2, \hat{d}$	$\hat{k}_1, \hat{d}_1$	$\hat{k}_2, \hat{d}_2$	$\hat{k}_3, \hat{d}_3$					
0.80	30	0.954434	0.745175	0.658705	0.397249	0.664927	0.556962	0.54976	0.491439	0.407458	0.369504					
	75	0.243572	0.225034	0.214071	0.193319	0.21396	0.197882	0.199969	0.19412	0.18977	0.171529					
	100	0.155654	0.147532	0.142172	0.134427	0.142246	0.134283	0.134366	0.131272	0.119391	0.112936					
	200	0.0596	0.058279	0.05791	0.056078	0.057784	0.056329	0.05707	0.054951	0.05526	0.053479					
	300	0.043403	0.042708	0.042526	0.041615	0.042386	0.041726	0.042125	0.040818	0.041629	0.040254					
0.85	400	0.027852	0.0276	0.027243	0.027175	0.027245	0.026866	0.026922	0.027116	0.026777	0.02671					
	30	1.946168	1.374725	1.105877	0.444471	1.142415	0.871689	0.811171	0.783915	0.47957	0.220401					
	75	0.204147	0.187363	0.170582	0.162854	0.171532	0.150601	0.153412	0.154754	0.133792	0.118013					
	100	0.216372	0.199915	0.18919	0.176707	0.189705	0.175818	0.172742	0.167071	0.148746	0.137642					
	200	0.063762	0.062281	0.06169	0.060056	0.061602	0.060182	0.060614	0.059024	0.059278	0.057201					
0.90	300	0.047064	0.04617	0.045327	0.044971	0.045353	0.043633	0.044212	0.04359	0.040982	0.040547					
	400	0.033515	0.033087	0.032715	0.032484	0.032717	0.032135	0.032206	0.032078	0.031235	0.031022					
	30	1.298845	1.001232	0.863607	0.493065	0.879004	0.723748	0.695048	0.649645	0.481287	0.403557					
	75	0.474685	0.404673	0.348511	0.300497	0.353658	0.298148	0.285015	0.304919	0.16321	0.165998					
	100	0.167196	0.157658	0.150745	0.144494	0.150997	0.140059	0.140765	0.136842	0.120605	0.114824					
0.95	200	0.082326	0.079812	0.078686	0.076566	0.078707	0.076181	0.076221	0.074165	0.070659	0.068255					
	300	0.069106	0.067121	0.06617	0.064595	0.066189	0.06416	0.064151	0.062649	0.059681	0.057877					
	400	0.043546	0.042833	0.042579	0.04195	0.042582	0.04194	0.04189	0.041208	0.040408	0.039632					
	30	0.872092	0.680739	0.566294	0.382719	0.572659	0.461666	0.457101	0.45648	0.318067	0.248441					
	75	0.888164	0.703876	0.596789	0.431621	0.615662	0.50021	0.470308	0.493122	0.284858	0.217953					
0.99	100	0.364473	0.317473	0.279857	0.254023	0.282832	0.241069	0.235827	0.249498	0.142692	0.148843					
	200	0.171198	0.159401	0.152903	0.144985	0.153305	0.142855	0.140909	0.131511	0.119641	0.111291					
	300	0.117352	0.111699	0.109148	0.105081	0.109275	0.104878	0.103395	0.098605	0.093937	0.089283					
	400	0.057152	0.055884	0.055325	0.054447	0.055335	0.054199	0.053989	0.053057	0.051324	0.050198					
	30	10.476181	5.992148	5.267037	0.990339	5.518712	4.120036	3.784811	2.958779	2.398839	0.4535					
	75	2.714658	1.843567	1.461783	0.507179	1.536959	1.154585	1.052303	1.019848	0.575067	0.181102					
	100	1.770967	1.233547	0.966388	0.474758	1.010864	0.766818	0.698566	0.76559	0.426366	0.175966					
	200	1.006986	0.781929	0.637725	0.448945	0.663145	0.520263	0.478881	0.532758	0.244835	0.169568					
	300	0.494804	0.419079	0.36784	0.318364	0.374655	0.316366	0.299991	0.314038	0.143268	0.163677					
	400	0.400952	0.350014	0.319306	0.282005	0.323309	0.283408	0.271218	0.25546	0.151138	0.168484					

**Table 2.** Estimated MSE values for biased estimators in PRM under multicollinearity when $p = 3$ and $\beta_0 = 0$.

ρ	n	PMLE		PRRE		PLRE		PMRTE		PARTE	
		-	$\hat{k}_1$	$\hat{k}_2$	$\hat{d}_1$	$\hat{d}_2$	$\hat{k}_1, \hat{d}$	$\hat{k}_2, \hat{d}$	$\hat{k}_1, \hat{d}_1$	$\hat{k}_2, \hat{d}_2$	$\hat{k}_3, \hat{d}_3$
0.80	30	0.324824	0.272111	0.243098	0.242149	0.250197	0.216014	0.201949	0.206967	0.124241	0.150022
	75	0.087145	0.08248	0.080536	0.081508	0.081508	0.07701	0.075991	0.073536	0.068356	0.069449
	100	0.056797	0.054788	0.054282	0.054527	0.054527	0.052848	0.052426	0.050612	0.049044	0.049376
	200	0.022459	0.022087	0.022009	0.02205	0.02205	0.021762	0.021667	0.021319	0.021083	0.021142
	300	0.015603	0.015426	0.01539	0.015411	0.015411	0.015212	0.015227	0.014898	0.014758	0.014799
0.85	400	0.01031	0.010238	0.010226	0.010233	0.010233	0.010162	0.010161	0.01004	0.009995	0.010008
	30	0.603959	0.464799	0.392004	0.36215	0.396937	0.33377	0.299022	0.316914	0.156546	0.165544
	75	0.072035	0.068065	0.067189	0.067437	0.067437	0.064707	0.063715	0.060755	0.058153	0.058371
	100	0.080128	0.075894	0.074724	0.075199	0.0752	0.071952	0.070845	0.068014	0.06461	0.065166
	200	0.023622	0.023241	0.023169	0.023209	0.023209	0.022853	0.022824	0.022274	0.022023	0.022091
0.90	300	0.017717	0.01748	0.017444	0.017465	0.017465	0.017211	0.017234	0.016759	0.016608	0.016648
	400	0.01249	0.012381	0.012366	0.012375	0.012375	0.012274	0.01227	0.012086	0.01203	0.012046
	30	0.440595	0.363616	0.309672	0.316333	0.325913	0.265833	0.249356	0.268575	0.150365	0.173373
	75	0.17139	0.152155	0.145654	0.146453	0.146606	0.134773	0.12944	0.123817	0.105941	0.107746
	100	0.062473	0.059904	0.059377	0.059589	0.05959	0.057631	0.057091	0.054665	0.052974	0.053247
0.95	200	0.030956	0.030287	0.030183	0.030241	0.030241	0.029718	0.029592	0.028797	0.028443	0.028523
	300	0.0252	0.024698	0.024616	0.024665	0.024665	0.024283	0.024169	0.023626	0.02336	0.023432
	400	0.015828	0.015645	0.015624	0.015637	0.015637	0.015505	0.015464	0.015239	0.01517	0.015191
	30	0.299941	0.246776	0.223664	0.221	0.225671	0.199705	0.18429	0.183237	0.102865	0.135386
	75	0.325885	0.270399	0.241126	0.243217	0.247306	0.209796	0.197523	0.199217	0.107497	0.136817
0.99	100	0.135891	0.122525	0.118357	0.119194	0.119219	0.110316	0.107013	0.101125	0.090457	0.09056
	200	0.063143	0.059942	0.059265	0.059539	0.059539	0.05722	0.056456	0.053831	0.051764	0.052045
	300	0.044869	0.043243	0.042971	0.043102	0.043102	0.041946	0.04155	0.040003	0.039128	0.039298
	400	0.022057	0.021695	0.021653	0.021677	0.021677	0.021412	0.021338	0.020884	0.020739	0.020777
	30	3.612893	2.147229	1.847965	0.483698	1.837717	1.496241	1.341286	1.136437	0.851483	0.235557
	75	0.984018	0.699246	0.556975	0.450438	0.566822	0.448321	0.402766	0.421863	0.211198	0.162722
	100	0.67029	0.490407	0.4078	0.360993	0.412337	0.334375	0.304383	0.311529	0.172944	0.147788
	200	0.371293	0.301344	0.264956	0.267004	0.27109	0.225162	0.210897	0.212644	0.101403	0.133283
	300	0.182194	0.159818	0.151376	0.153257	0.153311	0.137553	0.132263	0.124894	0.103048	0.105132
	400	0.148233	0.133112	0.127942	0.129509	0.129509	0.118322	0.114629	0.10787	0.094419	0.095655

Table 3. Estimated MSE values for biased estimators in PRM under multicollinearity when $p = 3$ and $\beta_0 = 1$.

ρ	n	PMLE	PRRE		PLRE		PMRTE		PARTE		
		-	$\hat{k}_1$	$\hat{k}_2$	$\hat{d}_1$	$\hat{d}_2$	$\hat{k}_1, \hat{d}$	$\hat{k}_2, \hat{d}$	$\hat{k}_1, \hat{d}_1$	$\hat{k}_2, \hat{d}_2$	$\hat{k}_3, \hat{d}_3$
0.80	30	0.117515	0.113224	0.111928	0.108466	0.112053	0.108119	0.10791	0.102907	0.098638	0.094223
	75	0.031585	0.031359	0.031298	0.031104	0.031299	0.031181	0.031081	0.030991	0.030836	0.030632
	100	0.020157	0.020081	0.02005	0.01999	0.02005	0.01999	0.019971	0.019915	0.019828	0.019749
	200	0.007812	0.007802	0.007796	0.007787	0.007796	0.007786	0.007784	0.007778	0.007762	0.007752
	300	0.005863	0.005857	0.005854	0.005849	0.005854	0.005847	0.005848	0.005841	0.005832	0.005825
0.85	400	0.003728	0.003725	0.003724	0.003722	0.003724	0.00372	0.003721	0.003716	0.003712	0.003709
	30	0.20628	0.191521	0.183183	0.174487	0.183834	0.172504	0.16785	0.155347	0.14536	0.13581
	75	0.027234	0.027045	0.026949	0.026833	0.02695	0.026729	0.026737	0.02651	0.026175	0.025996
	100	0.028398	0.028161	0.028082	0.027909	0.028083	0.027923	0.027839	0.027693	0.027465	0.027244
	200	0.008895	0.008879	0.008872	0.008858	0.008872	0.008847	0.008854	0.008819	0.00879	0.008768
0.90	300	0.006574	0.006562	0.006557	0.006549	0.006557	0.006539	0.006544	0.006515	0.006494	0.006477
	400	0.004558	0.004553	0.004551	0.004547	0.004551	0.004543	0.004545	0.004533	0.004524	0.004517
	30	0.153483	0.14672	0.145112	0.139192	0.145331	0.140947	0.13909	0.133769	0.129753	0.1231
	75	0.062249	0.060739	0.059972	0.059145	0.059992	0.058814	0.058269	0.057793	0.055588	0.054503
	100	0.022599	0.022457	0.0224	0.022302	0.0224	0.022272	0.022247	0.022115	0.021929	0.021791
0.95	200	0.011319	0.011279	0.011264	0.011236	0.011264	0.01123	0.011221	0.011184	0.011135	0.011095
	300	0.009462	0.009431	0.009421	0.009398	0.009421	0.009398	0.009389	0.009366	0.009334	0.009304
	400	0.005571	0.005561	0.005558	0.005551	0.005558	0.005548	0.005547	0.005537	0.005524	0.005514
	30	0.108944	0.104046	0.101108	0.098658	0.101234	0.095531	0.095679	0.090027	0.082366	0.079194
	75	0.115175	0.109697	0.107831	0.103658	0.108	0.10419	0.102544	0.09907	0.094353	0.089426
0.99	100	0.046818	0.045827	0.045404	0.044788	0.045416	0.044406	0.044338	0.04328	0.041799	0.040885
	200	0.023123	0.022898	0.02282	0.022665	0.022821	0.022646	0.022584	0.022401	0.022153	0.021937
	300	0.01636	0.01625	0.016212	0.016137	0.016212	0.016131	0.016096	0.016018	0.015899	0.015796
	400	0.007563	0.007542	0.007534	0.007521	0.007534	0.007516	0.007512	0.007491	0.007463	0.007442
	30	1.298725	0.931276	0.754834	0.438487	0.794446	0.612221	0.559073	0.606746	0.335096	0.178611
	75	0.359099	0.31629	0.292422	0.261469	0.295891	0.262678	0.251998	0.231397	0.173988	0.163999
	100	0.239273	0.215185	0.200524	0.186278	0.202002	0.183224	0.176644	0.170982	0.137546	0.127446
	200	0.129587	0.122305	0.11942	0.114139	0.119656	0.114193	0.112168	0.10398	0.100358	0.094245
	300	0.066755	0.064725	0.06394	0.06255	0.063973	0.062399	0.061823	0.060547	0.058152	0.056346
	400	0.051671	0.050442	0.050027	0.049143	0.050042	0.049124	0.04877	0.047879	0.046591	0.045397

**Table 4.** Estimated MSE values for biased estimators in PRM under multicollinearity when $p = 7$ and $\beta_0 = -1$.

ρ	n	PMLE		PRRE		PLRE		PMRTE		PARTE	
		-		$\hat{k}_1$	$\hat{k}_2$	$\hat{d}_1$	$\hat{d}_2$	$\hat{k}_1, \hat{d}$	$\hat{k}_2, \hat{d}$	$\hat{k}_1, \hat{d}_1$	$\hat{k}_2, \hat{d}_2$
0.80	30	2.025387	1.569177	1.357302	0.682814	0.356513	1.231137	1.095437	0.711053	0.812549	0.568669
	75	0.48907	0.448522	0.402595	0.377686	0.407083	0.380751	0.362152	0.336829	0.305986	0.293603
	100	0.169105	0.161635	0.156719	0.150736	0.156724	0.153634	0.149416	0.133264	0.149667	0.134056
	200	0.097532	0.095458	0.092061	0.092515	0.092713	0.090023	0.089193	0.088484	0.087078	0.085417
	300	0.035447	0.035202	0.034736	0.034862	0.034866	0.034157	0.034353	0.034062	0.034078	0.034152
0.85	400	0.028674	0.028489	0.028191	0.028249	0.028252	0.02777	0.027904	0.027477	0.027364	0.027234
	30	4.391535	1.432825	1.19101	0.627961	1.19783	1.078811	0.94413	0.63371	0.698492	0.473721
	75	0.401981	0.374196	0.341053	0.328637	0.344762	0.325716	0.31052	0.288621	0.264653	0.264306
	100	0.306772	0.285493	0.253473	0.25489	0.259538	0.236186	0.22478	0.221244	0.17772	0.176692
	200	0.06924	0.068168	0.066849	0.066781	0.066984	0.06542	0.065375	0.063362	0.062867	0.061408
0.90	300	0.038855	0.037944	0.038148	0.038144	0.038148	0.037663	0.037398	0.037585	0.037028	0.036816
	400	0.016181	0.016129	0.016068	0.016046	0.01605	0.01603	0.016031	0.016046	0.016181	0.016194
	30	2.928903	2.134333	1.725438	0.767273	1.747073	1.544598	1.320078	0.923933	0.994574	0.433131
	75	1.038168	0.873204	0.683043	0.591857	0.70491	0.610411	0.543514	0.531044	0.426665	0.291823
	100	0.401322	0.374564	0.331482	0.333988	0.341811	0.308874	0.298227	0.293474	0.251413	0.246433
0.95	200	0.096949	0.095015	0.091162	0.092492	0.092515	0.088646	0.087692	0.087635	0.08381	0.082409
	300	0.064093	0.063298	0.061955	0.062374	0.062374	0.060642	0.060434	0.059195	0.057502	0.056246
	400	0.036958	0.036686	0.036241	0.036366	0.036367	0.03595	0.035736	0.035597	0.035169	0.034787
	30	9.81437	6.998229	5.639281	1.139952	5.624675	5.056886	4.239809	2.739537	3.173423	0.677593
	75	0.856613	0.750385	0.596759	0.573036	0.61781	0.535631	0.481565	0.49295	0.363015	0.295145
0.99	100	0.605686	0.553417	0.491086	0.472309	0.496107	0.463027	0.426461	0.411483	0.316823	0.290545
	200	0.104931	0.102686	0.098378	0.099955	0.099956	0.094381	0.094138	0.091758	0.086257	0.084124
	300	0.120194	0.117509	0.113421	0.114378	0.114406	0.110759	0.108577	0.10715	0.103232	0.098891
	400	0.070676	0.069774	0.068736	0.068727	0.068813	0.067913	0.067317	0.065782	0.065596	0.06379
	30	32.555579	23.056949	18.819827	2.899639	18.781603	16.911246	14.082561	8.00158	10.366899	1.076249
	75	5.416385	4.14548	3.221123	1.280528	3.273119	2.903431	2.439634	1.915611	1.834974	0.544549
	100	2.505318	1.94909	1.502848	0.933959	1.516171	1.350257	1.146865	1.005936	0.878365	0.411985
	200	0.879217	0.783164	0.675374	0.626794	0.681653	0.62985	0.565786	0.5027	0.415737	0.329741
	300	0.599425	0.545332	0.492202	0.464499	0.494974	0.466904	0.429228	0.401954	0.320967	0.277657
	400	0.307066	0.290801	0.276722	0.269733	0.277106	0.268848	0.255862	0.230522	0.184031	0.209637

Table 5. Estimated MSE values for biased estimators in PRM under multicollinearity when $p = 7$ and $\beta_0 = 0$.

ρ	n	PMLE			PRRE			PLRE			PMRTE			PARTE		
		-	$\hat{k}_1$	$\hat{k}_2$	$\hat{d}_1$	$\hat{d}_2$	$\hat{k}_1, \hat{d}$	$\hat{k}_2, \hat{d}$	$\hat{k}_1, \hat{d}_1$	$\hat{k}_2, \hat{d}_2$	$\hat{k}_3, \hat{d}_3$					
0.80	30	0.570031	0.483947	0.432363	0.413618	0.437807	0.406531	0.362635	0.306889	0.266423	0.234193					
	75	0.165056	0.155404	0.151758	0.15282	0.15282	0.148388	0.142656	0.129288	0.115157	0.127609					
	100	0.061628	0.059887	0.059589	0.059691	0.059691	0.059004	0.058058	0.054349	0.05707	0.05534					
	200	0.034612	0.034126	0.034062	0.03409	0.03409	0.033909	0.033638	0.032527	0.033385	0.032932					
	300	0.01291	0.012851	0.012847	0.012849	0.012849	0.012825	0.012797	0.012625	0.012754	0.012695					
0.85	400	0.010541	0.010499	0.010497	0.010498	0.010498	0.010483	0.010462	0.010344	0.010435	0.010393					
	30	0.625697	0.507886	0.440203	0.416963	0.444704	0.407702	0.35496	0.293846	0.262416	0.214162					
	75	0.144574	0.137661	0.135559	0.136143	0.136143	0.133263	0.12908	0.117926	0.125634	0.118543					
	100	0.115493	0.10996	0.108787	0.10894	0.10894	0.107031	0.103943	0.094025	0.101228	0.095515					
	200	0.025199	0.024936	0.024912	0.024923	0.024923	0.024829	0.024688	0.024032	0.024545	0.024292					
0.90	300	0.014372	0.014298	0.014293	0.014296	0.014296	0.014269	0.014231	0.014034	0.014188	0.014118					
	400	0.005781	0.005768	0.005767	0.005768	0.005768	0.005763	0.005756	0.005721	0.005749	0.005737					
	30	0.954264	0.73046	0.612363	0.529054	0.616412	0.559831	0.479636	0.38369	0.367997	0.258113					
	75	0.345407	0.300121	0.278792	0.279063	0.280469	0.264719	0.242165	0.207422	0.186957	0.175162					
	100	0.145745	0.13865	0.136566	0.137105	0.137105	0.134067	0.129988	0.117975	0.125987	0.118708					
0.95	200	0.035586	0.035086	0.035039	0.035054	0.035054	0.034869	0.034614	0.033334	0.034304	0.033811					
	300	0.024241	0.024032	0.02402	0.024024	0.024024	0.02395	0.023846	0.023274	0.023717	0.023509					
	400	0.01325	0.013187	0.013184	0.013186	0.013186	0.013162	0.013132	0.012954	0.013091	0.01303					
	30	3.302476	2.420138	1.914207	1.06714	1.906811	1.730516	1.441345	1.09616	1.095657	0.437853					
	75	0.306518	0.276951	0.264029	0.265	0.265219	0.254073	0.236999	0.205334	0.171333	0.182503					
0.99	100	0.220982	0.206427	0.201209	0.202179	0.202179	0.196175	0.187592	0.16692	0.137246	0.163271					
	200	0.039973	0.039337	0.039277	0.039295	0.039295	0.039071	0.038738	0.037169	0.038377	0.037751					
	300	0.044116	0.043408	0.043342	0.043363	0.043363	0.043104	0.04274	0.040931	0.04231	0.041609					
	400	0.026427	0.026184	0.026166	0.026174	0.026174	0.026087	0.025961	0.025324	0.025821	0.025587					
	30	10.686678	7.771544	6.190026	1.340691	6.095624	5.588227	4.644533	3.226349	3.500016	0.59675					
	75	1.923816	1.506111	1.184127	0.955961	1.191441	1.074872	0.902783	0.800643	0.68404	0.401389					
	100	0.877585	0.699006	0.59117	0.549714	0.593252	0.543373	0.467539	0.395828	0.353588	0.269572					
	200	0.31411	0.285836	0.273633	0.274846	0.274976	0.264173	0.247397	0.215825	0.179869	0.197402					
	300	0.215801	0.200294	0.195237	0.195851	0.195851	0.190017	0.181246	0.15923	0.135411	0.154932					
	400	0.11223	0.107655	0.106841	0.106953	0.106953	0.105411	0.102829	0.093825	0.100581	0.095861					

**Table 6.** Estimated MSE values for biased estimators in PRM under multicollinearity when $p = 7$ and $\beta_0 = 1$.

ρ	n	PMLE		PRRE		PLRE		PMRTE		PARTE		
		-		$\hat{k}_1$	$\hat{k}_2$	$\hat{d}_1$	$\hat{d}_2$	$\hat{k}_1, \hat{d}$	$\hat{k}_2, \hat{d}$	$\hat{k}_1, \hat{d}_1$	$\hat{k}_2, \hat{d}_2$	$\hat{k}_3, \hat{d}_3$
0.80	30	0.194763	0.188641	0.185328	0.181644	0.185482	0.181251	0.178668	0.159543	0.166584	0.156753	0.156753
	75	0.060588	0.059994	0.059753	0.059334	0.059757	0.059297	0.059123	0.057224	0.058052	0.05645	0.05645
	100	0.022326	0.022258	0.022213	0.022175	0.022213	0.02215	0.02215	0.021952	0.021988	0.021828	0.021828
	200	0.012784	0.012767	0.012757	0.012747	0.012757	0.01274	0.012737	0.012707	0.012704	0.012678	0.012678
	300	0.00482	0.004819	0.004817	0.004816	0.004817	0.004814	0.004815	0.004816	0.00481	0.004812	0.004812
0.85	400	0.003809	0.003808	0.003807	0.003806	0.003807	0.003805	0.003806	0.003803	0.003801	0.003801	0.003801
	30	0.227704	0.215283	0.206964	0.20066	0.207427	0.199545	0.19337	0.174256	0.151596	0.157141	0.157141
	75	0.051941	0.051563	0.051383	0.051137	0.051385	0.051192	0.050962	0.050205	0.050596	0.049741	0.049741
	100	0.041559	0.041213	0.040971	0.040843	0.040972	0.040637	0.040526	0.038292	0.039726	0.038831	0.038831
	200	0.009151	0.009138	0.009131	0.009124	0.009131	0.009115	0.009116	0.009063	0.009077	0.009038	0.009038
0.90	300	0.005304	0.005301	0.0053	0.005298	0.0053	0.005296	0.005296	0.005288	0.005288	0.005282	0.005282
	400	0.002134	0.002133	0.002133	0.002132	0.002133	0.002132	0.002132	0.002134	0.002131	0.002133	0.002133
	30	0.34706	0.315589	0.293129	0.276344	0.295138	0.276804	0.261392	0.233894	0.199982	0.179235	0.179235
	75	0.128112	0.123554	0.120571	0.118481	0.12067	0.117053	0.11524	0.104503	0.10746	0.097814	0.097814
	100	0.052722	0.052272	0.05206	0.05178	0.052062	0.051737	0.051556	0.050338	0.05088	0.049751	0.049751
0.95	200	0.012855	0.012828	0.01281	0.012798	0.01281	0.012763	0.012777	0.012623	0.012652	0.012558	0.012558
	300	0.008946	0.008932	0.008924	0.008919	0.008924	0.008903	0.008906	0.008836	0.008852	0.008803	0.008803
	400	0.00495	0.004947	0.004945	0.004943	0.004945	0.004942	0.00494	0.004932	0.004935	0.004926	0.004926
	30	1.153887	0.974008	0.81104	0.68882	0.826677	0.750035	0.655442	0.61862	0.507387	0.355476	0.355476
	75	0.113955	0.111317	0.109691	0.108365	0.109723	0.108032	0.106507	0.094799	0.103074	0.097032	0.097032
0.99	100	0.082394	0.081217	0.080678	0.079936	0.080689	0.080126	0.079363	0.077015	0.078335	0.0756	0.0756
	200	0.014099	0.014064	0.014041	0.014026	0.014041	0.01397	0.013997	0.013746	0.013798	0.013646	0.013646
	300	0.016005	0.01596	0.015936	0.015914	0.015936	0.015903	0.015881	0.015749	0.015805	0.015685	0.015685
	400	0.009673	0.009659	0.009651	0.009644	0.009651	0.009641	0.009635	0.009594	0.009611	0.009574	0.009574
	30	3.774225	2.978506	2.301757	1.255068	2.343354	2.080639	1.747728	1.541037	1.334532	0.513317	0.513317
	75	0.711556	0.637476	0.577957	0.525861	0.584359	0.548893	0.500902	0.440776	0.374275	0.295938	0.295938
	100	0.333973	0.308509	0.287115	0.27619	0.288188	0.274822	0.257911	0.236407	0.191826	0.183036	0.183036
	200	0.117703	0.114999	0.113503	0.111976	0.113537	0.112264	0.110334	0.095457	0.108186	0.102145	0.102145
	300	0.079588	0.078233	0.077485	0.076762	0.077497	0.076841	0.075878	0.067033	0.074723	0.071721	0.071721
	400	0.042341	0.04198	0.041766	0.041602	0.041767	0.041599	0.041315	0.040716	0.041029	0.040255	0.040255

Table 7. Estimated MSE values for biased estimators in PRM under multicollinearity when $p = 12$ and $\beta_0 = -1$.

ρ	n	PMLE		PRRE		PLRE		PMRTE		PARTE	
		-	$\hat{k}_1$	$\hat{k}_2$	$\hat{d}_1$	$\hat{d}_2$	$\hat{k}_1, \hat{d}$	$\hat{k}_2, \hat{d}$	$\hat{k}_1, \hat{d}_1$	$\hat{k}_2, \hat{d}_2$	$\hat{k}_3, \hat{d}_3$
0.80	30	6.526303	4.4207	3.800013	1.035546	3.746236	3.454666	2.852239	1.140431	2.235665	0.771904
	75	1.003581	0.877384	0.725331	0.639929	0.742469	0.674065	0.605829	0.491676	0.51531	0.381452
	100	0.224299	0.216932	0.208177	0.205413	0.208844	0.20451	0.200101	0.184674	0.196042	0.188739
	200	0.0416	0.04126	0.04062	0.040835	0.040835	0.040368	0.039981	0.03965	0.039631	0.038921
	300	0.021488	0.021397	0.021308	0.021284	0.021307	0.021257	0.021185	0.020791	0.021108	0.020871
0.85	400	0.01217	0.012138	0.01208	0.012095	0.012095	0.012046	0.012026	0.012019	0.011978	0.011958
	30	17.626034	10.613025	9.114177	1.269542	9.096516	8.197099	6.546195	2.421576	5.216058	1.165189
	75	0.371573	0.35111	0.317662	0.319997	0.324354	0.306127	0.289795	0.272404	0.258911	0.24422
	100	0.116204	0.113711	0.109172	0.110354	0.110421	0.106896	0.105065	0.10272	0.098903	0.098959
	200	0.064866	0.064277	0.063238	0.063572	0.063573	0.06256	0.062108	0.060615	0.060817	0.059285
0.90	300	0.031515	0.031316	0.03099	0.031086	0.031086	0.030677	0.030614	0.029754	0.029916	0.029262
	400	0.012729	0.012703	0.012672	0.012668	0.012668	0.012639	0.012637	0.012611	0.012605	0.012614
	30	24.079871	12.038083	10.202013	1.522363	10.568237	9.061669	7.12764	3.161241	5.87988	1.292361
	75	0.847007	0.774671	0.650964	0.643627	0.675483	0.613894	0.556625	0.503422	0.467124	0.39085
	100	1.245337	1.106493	0.905267	0.82074	0.931027	0.844366	0.749857	0.620067	0.613527	0.442779
0.95	200	0.109385	0.107439	0.103934	0.105156	0.105157	0.102073	0.100068	0.095733	0.085936	0.090638
	300	0.034301	0.034052	0.03363	0.033771	0.033771	0.033349	0.033143	0.03237	0.032572	0.031775
	400	0.028624	0.028536	0.02848	0.02842	0.028442	0.02843	0.028398	0.028004	0.02834	0.028176
	30	29.6533	21.678788	17.486621	1.586104	17.230246	16.025864	13.254022	6.297672	10.872633	1.129518
	75	3.253494	2.661461	2.068252	1.257276	2.085103	1.895086	1.614846	1.116266	1.327786	0.568171
0.99	100	0.894471	0.787945	0.6326	0.610832	0.649227	0.588169	0.512942	0.450492	0.428226	0.298114
	200	0.475912	0.452422	0.39897	0.417738	0.417738	0.381859	0.352883	0.345006	0.291249	0.270988
	300	0.043731	0.043438	0.042963	0.043111	0.043111	0.042673	0.0424	0.04154	0.041831	0.040898
	400	0.039739	0.039485	0.039067	0.039212	0.039212	0.038833	0.038558	0.037812	0.038104	0.037237
	30	62.902477	37.546632	32.650365	5.416655	32.703496	29.506294	23.64432	8.119852	19.082456	2.010048
	75	9.10695	7.136531	5.487746	2.787172	5.418119	5.042705	4.194969	2.588945	3.46701	0.765029
	100	3.698254	3.06929	2.304665	1.512065	2.308837	2.117332	1.769938	1.373128	1.465277	0.581126
	200	0.391389	0.372064	0.334825	0.345532	0.345697	0.322175	0.300251	0.285086	0.253737	0.232534
	300	0.248614	0.240516	0.228079	0.230342	0.230356	0.22334	0.214076	0.201221	0.183252	0.176914
	400	0.265213	0.2569	0.246215	0.246401	0.246735	0.241825	0.232692	0.215774	0.190715	0.195372

**Table 8.** Estimated MSE values for biased estimators in PRM under multicollinearity when $p = 12$ and $\beta_0 = 0$.

ρ	n	PMLE		PRRE		PLRE		PMRTE		PARTE	
		-	$\hat{k}_1$	$\hat{k}_2$	$\hat{d}_1$	$\hat{d}_2$	$\hat{k}_1, \hat{d}$	$\hat{k}_2, \hat{d}$	$\hat{k}_1, \hat{d}_1$	$\hat{k}_2, \hat{d}_2$	$\hat{k}_3, \hat{d}_3$
0.80	30	1.498765	1.180781	0.981258	0.780765	0.984638	0.913064	0.772757	0.478672	0.634195	0.343784
	75	0.348357	0.316083	0.30036	0.301004	0.301633	0.291216	0.271028	0.207177	0.292278	0.197207
	100	0.080376	0.078647	0.078323	0.078426	0.078426	0.078426	0.076763	0.069928	0.076135	0.072993
	200	0.014886	0.01481	0.014806	0.014808	0.014808	0.014784	0.014743	0.014355	0.014708	0.014571
	300	0.007882	0.007861	0.00786	0.007861	0.007861	0.007855	0.007843	0.00774	0.007836	0.0078
0.85	400	0.004527	0.004521	0.004521	0.004521	0.004521	0.004519	0.004516	0.004485	0.004513	0.004503
	30	3.543506	2.589218	2.110115	1.094571	2.095419	1.938004	1.599069	0.852981	1.297332	0.473095
	75	0.136124	0.130638	0.129306	0.129528	0.129528	0.127829	0.124392	0.106541	0.106701	0.111034
	100	0.04316	0.042498	0.042434	0.042449	0.042449	0.042271	0.041875	0.039094	0.041668	0.040493
	200	0.024517	0.024361	0.02435	0.024355	0.024355	0.024309	0.024219	0.023472	0.02416	0.023884
0.90	300	0.012006	0.011961	0.011959	0.01196	0.01196	0.011946	0.011921	0.011685	0.011901	0.011818
	400	0.004614	0.004608	0.004608	0.004608	0.004608	0.004606	0.004603	0.00457	0.0046	0.004589
	30	6.00064	3.985065	3.288176	1.282293	3.234382	3.004104	2.453328	1.235336	2.019009	0.615093
	75	0.301624	0.282075	0.273715	0.274598	0.274661	0.268074	0.254686	0.204098	0.207061	0.196999
	100	0.423341	0.387587	0.367788	0.3691	0.369652	0.357118	0.332265	0.256991	0.268958	0.229279
0.95	200	0.040094	0.039599	0.039559	0.039569	0.039569	0.039435	0.039142	0.036925	0.038976	0.038107
	300	0.012242	0.012182	0.01218	0.012181	0.012181	0.012165	0.012131	0.01184	0.01211	0.012006
	400	0.0105	0.010475	0.010473	0.010474	0.010474	0.010467	0.010453	0.010333	0.010445	0.010402
	30	9.312832	6.962979	5.513505	1.507016	5.432795	5.073574	4.195881	2.31251	3.467337	0.728447
	75	1.060158	0.89488	0.757602	0.731231	0.764095	0.711123	0.614224	0.451659	0.50795	0.343962
0.99	100	0.321239	0.290238	0.277126	0.277359	0.277657	0.268858	0.249653	0.18606	0.208492	0.174609
	200	0.172954	0.166515	0.164921	0.165136	0.165136	0.163165	0.15897	0.136379	0.146734	0.143253
	300	0.016033	0.015953	0.01595	0.015951	0.015951	0.015929	0.015884	0.015483	0.015853	0.01571
	400	0.0142	0.014137	0.014134	0.014135	0.014135	0.014118	0.014082	0.013762	0.014058	0.013945
	30	13.939435	9.66343	8.07939	2.515278	7.900118	7.43157	6.151548	2.790033	5.085548	0.870365
	75	3.291761	2.631575	2.060193	1.41993	2.051543	1.902649	1.589923	1.080094	1.316474	0.551297
	100	1.338193	1.131265	0.942472	0.897629	0.944991	0.882211	0.757383	0.569387	0.629197	0.403535
	200	0.14616	0.14057	0.139343	0.13947	0.13947	0.137846	0.134327	0.114854	0.113159	0.120856
	300	0.093911	0.09161	0.091281	0.091318	0.091318	0.090698	0.089287	0.08026	0.088558	0.084159
	400	0.097673	0.09534	0.09502	0.095051	0.095051	0.094423	0.092989	0.083497	0.092227	0.087821

Table 9. Estimated MSE values for biased estimators in PRM under multicollinearity when $p = 12$ and $\beta_0 = 1$.

ρ	n	PMLE		PRRE		PLRE		PMRTE		PARTE	
		-	$\hat{k}_1$	$\hat{k}_2$	$\hat{d}_1$	$\hat{d}_2$	$\hat{k}_1, \hat{d}$	$\hat{k}_2, \hat{d}$	$\hat{k}_1, \hat{d}_1$	$\hat{k}_2, \hat{d}_2$	$\hat{k}_3, \hat{d}_3$
0.80	30	0.497067	0.457032	0.424185	0.404282	0.426549	0.40681	0.381227	0.311751	0.325491	0.262895
	75	0.124468	0.121955	0.120077	0.119128	0.120106	0.11865	0.116903	0.106408	0.110743	0.106155
	100	0.029499	0.029435	0.029387	0.029353	0.029387	0.029299	0.029308	0.028987	0.029098	0.02887
	200	0.005524	0.005522	0.00552	0.005519	0.00552	0.005518	0.005518	0.005512	0.005513	0.005507
	300	0.002865	0.002864	0.002864	0.002863	0.002864	0.002862	0.002863	0.002856	0.002859	0.002854
0.85	400	0.001691	0.001691	0.00169	0.00169	0.00169	0.00169	0.00169	0.001692	0.00169	0.001691
	30	1.029314	0.896424	0.778568	0.683944	0.789652	0.729396	0.653194	0.503773	0.544793	0.36202
	75	0.051105	0.050739	0.05046	0.050332	0.050461	0.050247	0.049977	0.047436	0.049584	0.048333
	100	0.016	0.015962	0.015933	0.015919	0.015933	0.01589	0.015882	0.01567	0.015784	0.015633
	200	0.008835	0.008827	0.00882	0.008817	0.00882	0.00881	0.008808	0.008746	0.008783	0.00874
0.90	300	0.004418	0.004416	0.004414	0.004414	0.004414	0.004411	0.004411	0.004394	0.004403	0.004392
	400	0.001738	0.001738	0.001738	0.001738	0.001738	0.001737	0.001737	0.001737	0.001736	0.001736
	30	1.744772	1.432303	1.180608	0.895146	1.197144	1.098418	0.949415	0.699304	0.807001	0.444521
	75	0.108212	0.106698	0.105703	0.104995	0.105714	0.104916	0.103808	0.092201	0.102381	0.096945
	100	0.15413	0.151162	0.149107	0.147787	0.149136	0.147658	0.14535	0.127867	0.142897	0.132428
0.95	200	0.014793	0.014763	0.014743	0.014731	0.014743	0.014707	0.014704	0.014442	0.014608	0.014432
	300	0.004663	0.004659	0.004657	0.004655	0.004657	0.004652	0.004652	0.004623	0.00464	0.004621
	400	0.003811	0.00381	0.003809	0.003809	0.003809	0.003808	0.003808	0.003807	0.003806	0.003805
	30	3.2644	2.645296	2.031597	1.019484	2.054773	1.872782	1.569391	1.141374	1.299819	0.541938
	75	0.380231	0.359795	0.344259	0.33283	0.345024	0.335689	0.320019	0.258969	0.26771	0.232946
0.99	100	0.116593	0.113712	0.1115	0.110493	0.111532	0.109821	0.107751	0.095383	0.097796	0.094468
	200	0.062783	0.062307	0.062014	0.061799	0.062016	0.061789	0.061414	0.055318	0.061025	0.059267
	300	0.005906	0.005901	0.005898	0.005896	0.005898	0.005894	0.005892	0.005862	0.005882	0.00586
	400	0.005337	0.005333	0.00533	0.005329	0.00533	0.005328	0.005325	0.005306	0.00532	0.005305
	30	4.455907	3.387959	2.678865	1.392706	2.669415	2.456826	2.06935	1.313966	1.743938	0.583184
	75	1.158081	1.029325	0.863227	0.802229	0.869634	0.813402	0.71451	0.611718	0.59752	0.400734
	100	0.485098	0.455321	0.426072	0.413315	0.427111	0.414064	0.387857	0.32333	0.323677	0.263474
	200	0.055421	0.054959	0.054671	0.054469	0.054673	0.054455	0.054088	0.04878	0.053719	0.052018
	300	0.033956	0.033791	0.033684	0.033621	0.033684	0.033609	0.03347	0.031104	0.033348	0.032753
	400	0.035331	0.035162	0.035046	0.034986	0.035047	0.034971	0.034822	0.034143	0.034704	0.034095

nuisance parameter; it also dictates the effective shape of the Poisson log-likelihood and, hence, the effectiveness of each of these biased estimators in reducing MSE compared to the PMLE. While the gain of the proposed PARTE estimator over existing competitors is consistent across all three parameters, it is largest when the conditions are most difficult: the baseline count is low, the sample size is small, and multicollinearity is severe.

Under most conditions, particularly for severe multicollinearity ($\rho = 0.95, 0.99$) and low sample sizes ($n = 30, 75$), the proposed biased estimators (PRRE, PLRE, PMRTE, and PARTE) have lower MSE than the PMLE. This shows that they are successful shrinkage estimators, with a slight bias allowing a substantial reduction in variance and thus more reliable estimation.

Among the competitors, the PARTE estimator, especially the one with optimal tuning parameters ($\hat{k}_3, \hat{d}_3$), consistently has the smallest MSE under the most difficult circumstances. This is especially the case in the presence of high collinearity and p . For instance, in Table 7 ($p = 12, \beta_0 = -1, \rho = 0.99, n = 30$), the MSE for PARTE($\hat{k}_3, \hat{d}_3$) is 2.010, a drastic improvement over the PMLE's MSE of 62.902. The extra flexibility in a two-parameter shrinkage scheme appears to be useful for high multicollinearity.

In summary, this simulation provides strong evidence that in estimating parameters of a PRM with multicollinearity, the PMLE can be very inefficient. The proposed biased estimators, particularly the PARTE, represent a promising solution. They offer a substantial gain in estimation efficiency for small to moderate sample sizes and become crucial as multicollinearity increases and the model becomes more complex. This suggests that such shrinkage estimators should be widely used in practice for correlated predictors.

4.3. Relative efficiency

We can also use simulation to calculate the relative efficiency, providing an alternative comparison for biased estimators. It offers a way to compare the performance of one estimator to another. The estimates in Tables 1–9 are taken as the benchmark to compute relative efficiency. Relative efficiency can be calculated using the following formula:

$$RE(\hat{\beta}_*) = \frac{MSE(\hat{\beta}^{PMLE})}{MSE(\hat{\beta}_*)}, \quad (45)$$

where $\hat{\beta}_*$ denotes either the PRRE, PLRE, PMRTE, or PARTE.

The relative efficiency analysis, plotted across varying sample sizes, multicollinearity levels, number of explanatory variables, and intercept values in Figures 1–4, consistently shows that the proposed PARTE outperforms all other estimators. Specifically, the PARTE variant $\hat{k}_3, \hat{d}_3$ achieves the highest RE in every scenario, confirming its superior precision and making it the unequivocally preferred choice for parameter estimation under the conditions studied.

5. Real data applications

This section presents two real-world applications that utilize actual data to illustrate the scientific and practical benefits of the proposed estimator compared to current estimators based on MSE.

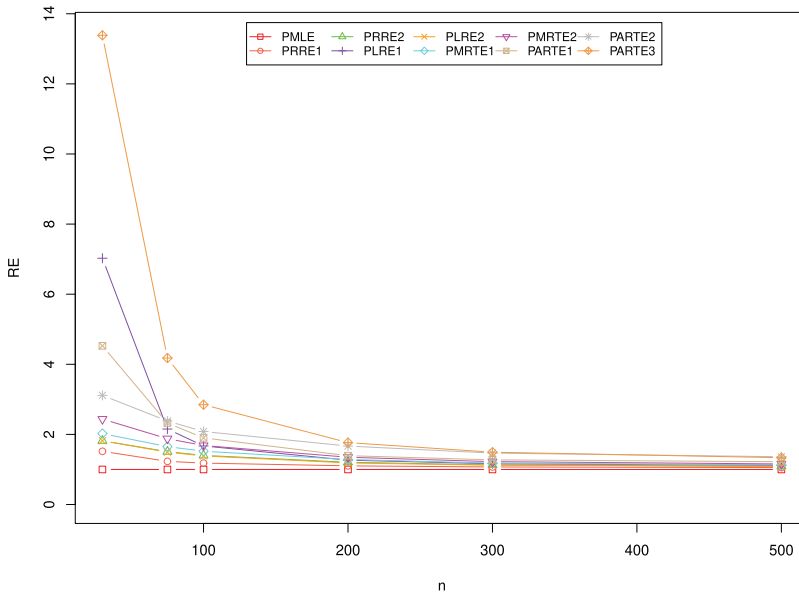


Figure 1. RE with sample size for estimators in the PRM.

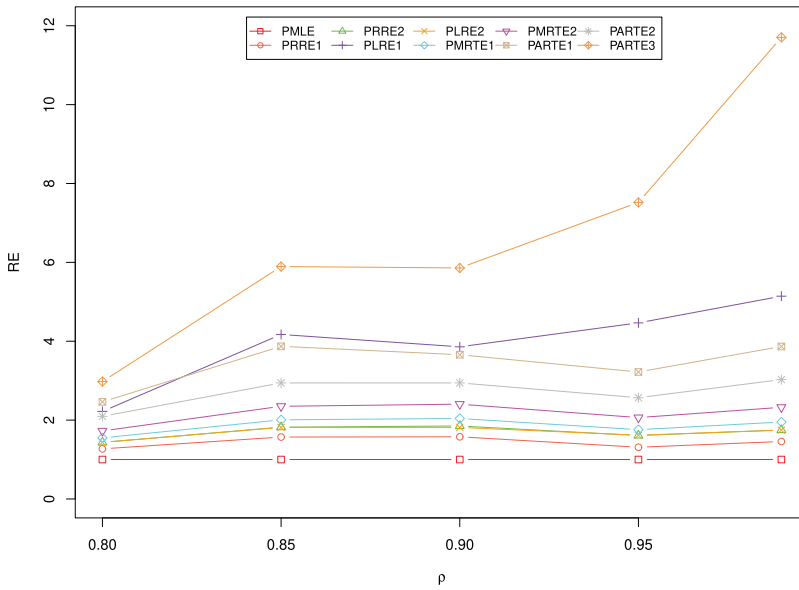


Figure 2. RE with the level of multicollinearity for estimators in the PRM.

5.1. League baseball dataset

To evaluate the performance of the proposed PARTE model in addressing the multicollinearity problem in the PRM model, we use a real-world dataset from Major League Baseball (MLB) teams in the United States during 2016. This dataset is available from Montgomery et al. [35]. It comprises 30 baseball teams, including 30 observations and 11

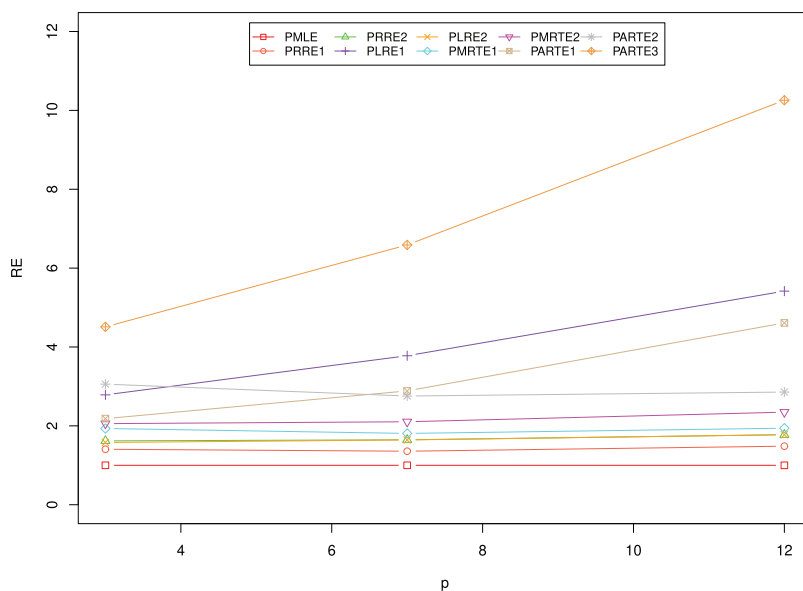


Figure 3. RE with the number of explanatory variables for estimators in the PRM.

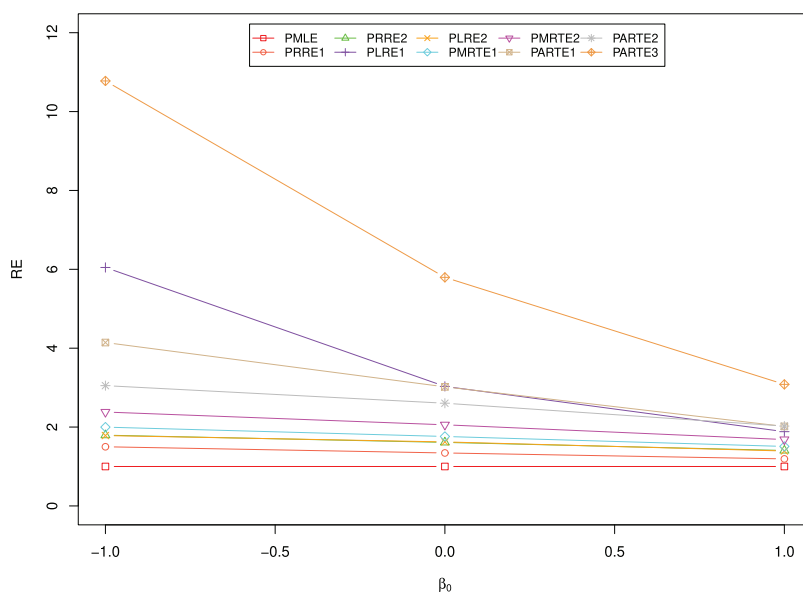


Figure 4. RE with the intercept value for estimators in the PRM.

variables. We analyze the factors affecting the number of wins for each team; therefore, we use the number of wins as the dependent variable (y), which is modelled with ten independent variables: average batter age (x_1), total runs (x_2), total home runs (x_3), slugging percentage (x_4), average pitcher age (x_5), average earned run (x_6), strikeouts (x_7), home runs allowed (x_8), runs allowed per game (x_9), errors (x_{10}).

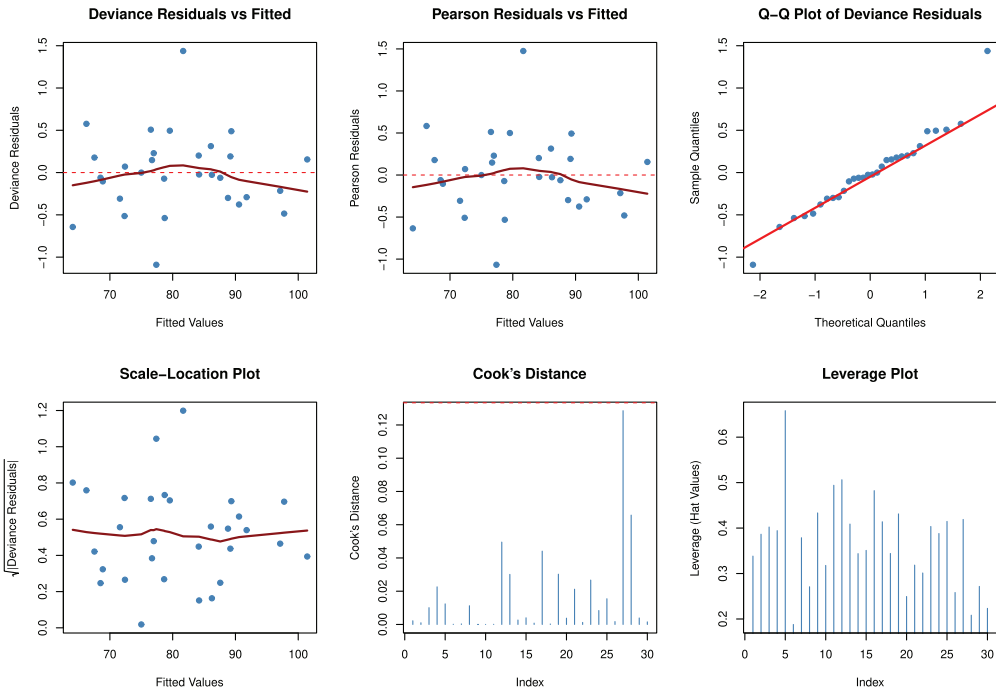


Figure 5. Diagnostic plots for the fitted PRM in the Major League Baseball dataset.

To verify that the PRM is the best choice of model for this data, a series of diagnostic checks was undertaken. The residual deviance of 6.32 on 19 degrees of freedom yields a chi-square p -value of 0.997, which is far above the p -value threshold of 0.05, indicating that the model fits the data very well. Both the AIC (215.06) and BIC (230.48) were lower than their counterparts for the negative binomial alternative (AIC = 217.06, BIC = 233.88), confirming the previous choice of the Poisson alternative over the negative binomial alternative. As for overdispersion, all three tests pointed in the same direction: the ratio of the deviance to the degrees of freedom was 0.333, the estimate of the dispersion using the Pearson statistic was 0.336, and the formal test for overdispersion produced $z = -10.293$ and $p = 1$, all of which confirm that there is no overdispersion and that the equidispersion assumption holds. The residual plots in Figure 5 confirm this: the deviance and Pearson residuals are randomly distributed around zero with no obvious pattern, the Q-Q plot shows that the residuals are close to the expected distribution, and the scale-location plot shows that the spread of the residuals is constant and stable across the range of fitted values. While the leverage plot shows that several points have somewhat higher leverage (this is to be expected in a sample of only $n = 30$ with ten predictors), none of the points have leverage greater than the Cook's distance cut-off of $4/n$, showing that no single observation has a large influence on the fitted model. In all, the diagnostics support the PRM as a good fit to the data and that the model assumptions are reasonably met.

We first check for multicollinearity between the explanatory variables. This is done using the correlation matrix (CM), the condition number (CN), and the variance inflation factor (VIF). The CM between the explanatory variables is presented in Figure 6, and it

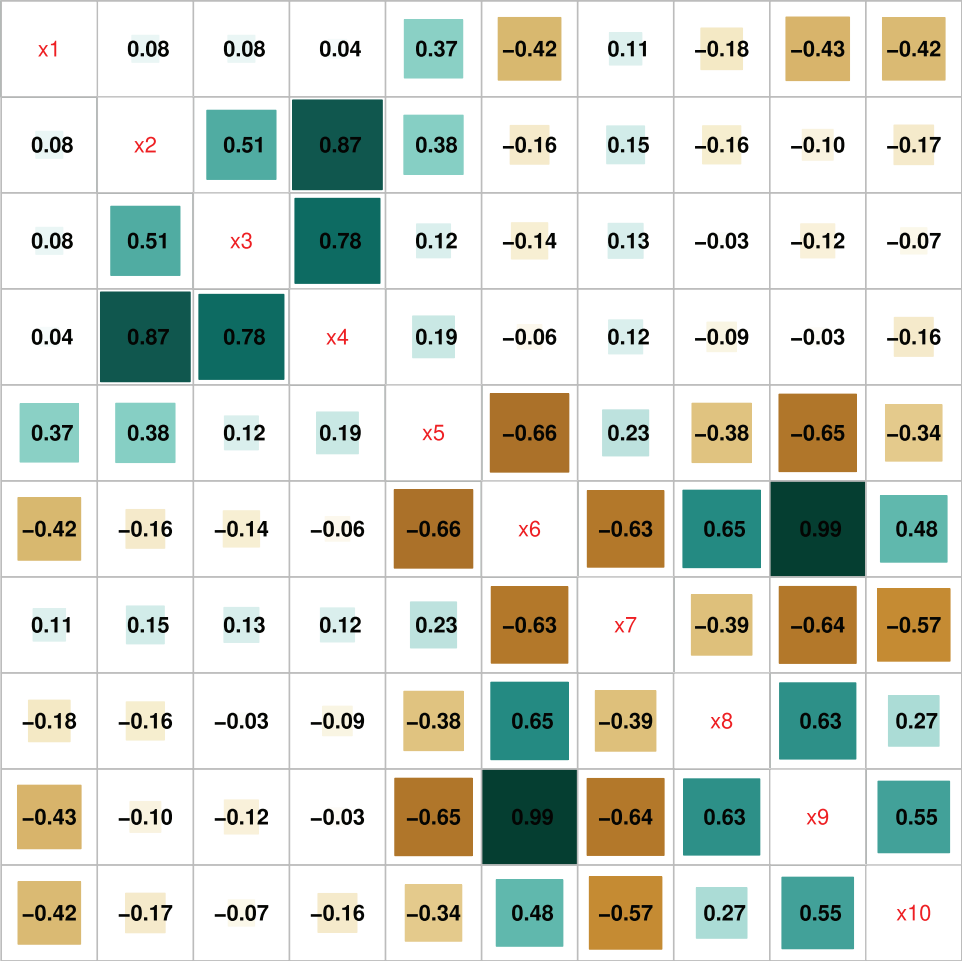


Figure 6. Correlation matrix between explanatory variables in the Major League Baseball dataset.

shows high correlations. The CN of 311273.8 is much larger than 30, also indicating multicollinearity. The VIFs for the explanatory variables are 1.55, 10.10, 4.47, 14.94, 2.91, 100.37, 2.93, 1.94, 97.56, and 3.32, all clearly exceeding 10, thus confirming strong multicollinearity among the explanatory variables.

Table 10 details the estimated regression coefficients and the corresponding MSE for the various estimators applied to the Major League Baseball dataset. This empirical application assesses the efficacy of the proposed methods in a practical setting characterized by substantial multicollinearity among predictors.

The most immediate observation is the substantial disparity in performance between the traditional PMLE and the biased estimators. The PMLE has the highest MSE of all the methods tested, at about 20.88. This elevated error rate suggests that the PMLE estimates are unstable and possess high variance, a known consequence of multicollinearity.

In contrast, all shrinkage methods, including the PRRE, PLRE, and the proposed estimators, demonstrate superior efficiency. The PARTE class of estimators consistently

Table 10. Estimated coefficients and MSEs of different estimators in the Major League Baseball dataset.

Estimator	Parameter	beta_0	beta_1	beta_2	beta_3	beta_4	beta_5	beta_6	beta_7	beta_8	beta_9	beta_10	MSE
PMLE	$\hat{\alpha}$	4.700944	0.003762	0.000794	0.000379	-0.05407	0.00125	-0.153047	-8.9e-05	0.000115	-0.034096	-0.002051	20.875522
	$\hat{\alpha}_1$	3.932516	0.009011	0.000682	0.000266	0.400272	0.010624	-0.104668	8e-06	8.1e-05	-0.056593	-0.001569	9.101102
PRRE	$\hat{\alpha}_2$	3.854882	0.009583	0.000675	0.000261	0.424538	0.0116	-0.098595	1.8e-05	7.5e-05	-0.059669	-0.001517	8.581985
	$\hat{\alpha}_1$	1.013288	0.03247	0.000667	0.000396	0.257073	0.049438	0.137623	0.000438	-0.00022	-0.167697	0.000341	14.211562
PLRE	$\hat{\alpha}_2$	3.993282	0.009271	0.00077	0.000382	0.005638	0.010497	-0.097267	1.2e-05	5.1e-05	-0.059734	-0.001592	14.762214
	$\hat{\alpha}_1, \hat{\alpha}$	3.782168	0.010124	0.00067	0.000257	0.444531	0.012518	-0.092766	2.8e-05	6.9e-05	-0.062643	-0.001469	8.164076
PMRTE	$\hat{\alpha}_2, \hat{\alpha}$	3.854882	0.009583	0.000675	0.000261	0.424538	0.0116	-0.098595	1.8e-05	7.5e-05	-0.059669	-0.001517	8.581985
	$\hat{\alpha}_1, \hat{\alpha}_1$	2.594492	0.019554	0.000656	0.000283	0.459561	0.027989	0.015441	0.000201	-4.9e-05	-0.118734	-0.000655	7.266676
PARTE	$\hat{\alpha}_2, \hat{\alpha}_2$	3.354902	0.013432	0.000655	0.000253	0.494967	0.018006	-0.055204	8.9e-05	3.1e-05	-0.082248	-0.001178	6.895672
	$\hat{\alpha}_3, \hat{\alpha}_3$	3.271991	0.014087	0.000654	0.000254	0.497749	0.019081	-0.047631	0.000101	2.3e-05	-0.086213	-0.001121	6.798453

outperforms the others. The optimal variant, defined by the parameters, achieves a minimum MSE of 6.80. This progression suggests that the additional flexibility provided by the two-parameter structure enables a more refined management of the bias-variance trade-off than single-parameter approaches, such as PRRE or PLRE.

Beyond error metrics, the behaviour of the regression coefficients provides insight into the mechanics of these estimators. While the direction (sign) of the relationships generally remains consistent across methods, the magnitude of the coefficients varies. The PMLE coefficients are larger in absolute value for several predictors. The biased estimators induce a shrinkage effect, pulling these coefficients towards zero. This shrinkage is particularly evident in the PARTE results, suggesting that this method effectively mitigates the variance inflation caused by correlated predictors, leading to more stable and reliable parameter estimates.

The analysis of the Major League Baseball dataset strongly corroborates the theoretical and simulation results. The proposed PARTE estimator, particularly the formulation, proves to be the most robust method in this context. It successfully addresses the ill-conditioning of the data, offering a significant improvement in estimation precision over both the standard PMLE and existing biased estimators like the PRRE and PLRE.

5.2. Recreation demand dataset

To further assess the effectiveness of the proposed estimator in handling multicollinearity across different data types, we consider a second application. This dataset comes from recreation demand studies and contains information on the number of trips made by recreational boats to Lake Somerville in East Texas during 1980. The data are available from Cameron and Trivedi [1] and were previously used by Abdelwahab et al. [8]. The sample consists of 179 observations and four variables. The primary outcome of interest is the number of trips, taken as the dependent variable (y), which is modelled using three explanatory variables: expenditure in US dollars to visit Lake Conroe (x_1), expenditure in US dollars to visit Lake Somerville (x_2), and expenditure in US dollars to visit Lake Houston (x_3).

To check whether the fitted PRM is appropriate, we performed a full set of diagnostic tests, covering goodness-of-fit, overdispersion, residual behaviour, and influential observations. The residual deviance of 200.05 on 175 degrees of freedom gave a chi-square p-value of 0.094, which is above the 0.05 threshold, indicating that the model fits the data reasonably well. The AIC and BIC values of 683.05 and 695.80 were both lower than those of the negative binomial alternative (AIC = 685.05, BIC = 700.99), which adds further support for choosing the Poisson specification. On the question of overdispersion, the deviance-to-degrees-of-freedom ratio of 1.143 and the Pearson dispersion estimate of 1.002 were both close to one, and a formal overdispersion test gave $z = -0.182$ with $p = 0.572$, all of which confirm that the equidispersion assumption holds comfortably. The residual plots in Figure 7 told a similar story: the deviance and Pearson residuals showed no clear patterns against the fitted values, the smoothed trend line stayed near zero throughout, and the Q-Q plot showed that the residuals follow the expected distribution closely, with only slight departures at the tails. The scale-location plot showed no notable change in residual spread across the range of fitted values. Overall, the diagnostic results confirm that the PRM fits the data well and that its underlying assumptions are adequately satisfied.

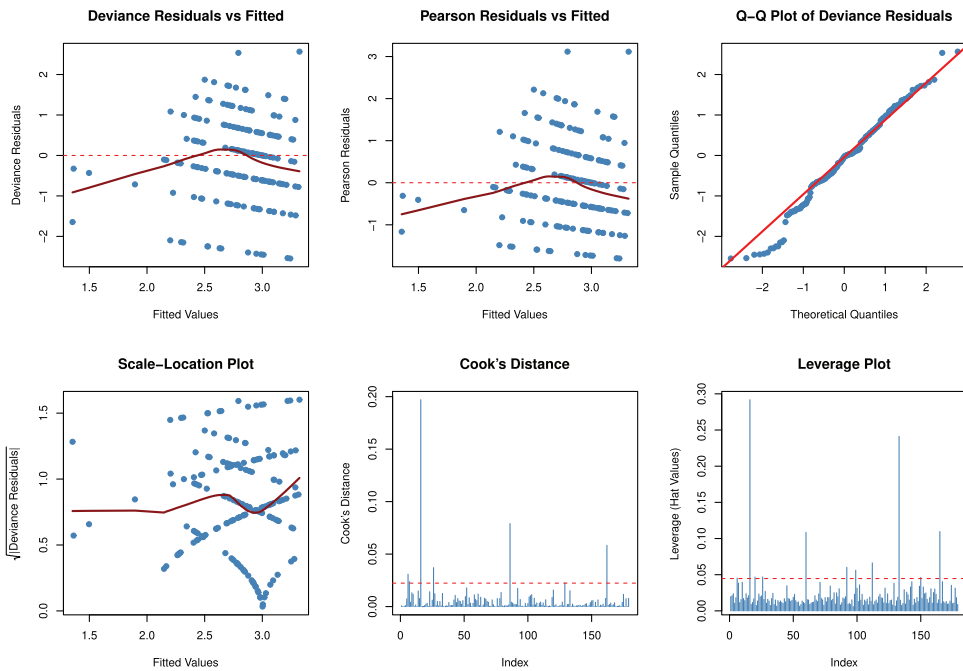


Figure 7. Diagnostic plots for the fitted PRM in the recreation demand dataset.

Before modelling the data, we verify the presence of multicollinearity among the explanatory variables. Three main measures are used for this verification: CM, CN, and VIF. Figure 8 shows the two-way CM among the explanatory variables, which demonstrates strong correlations. The CN is 33.77, significantly exceeding 30, further supporting the presence of multicollinearity. The VIFs for the explanatory variables are 43.84, 14.62, and 24.64, all clearly exceeding 10, thus confirming strong multicollinearity among the explanatory variables.

Table 11 shows the performance of the standard (PMLE) and proposed biased estimators for the recreation demand data. The findings show that the proposed estimators are more robust to multicollinearity than the existing estimators. The high correlation in the data results in the poor performance of the standard PMLE, with the largest MSE (1.224). All the regularized estimators (PRRE, PLRE, PMRTE, and PARTE) greatly improve this. The PARTE estimator is the best. It has an MSE of 0.491. This indicates that it's important to select the correct shrinkage parameters.

These findings are in line with the previous results. They show that when there is multicollinearity in the data, the new estimators, and in particular PARTE, outperform the standard methods. They give much smaller error rates and more accurate coefficient estimates.

6. Conclusion

This paper proposes a novel biased estimating method based on two shrinkage parameters to mitigate the effects of multicollinearity in PRMs. This method aims to reduce the

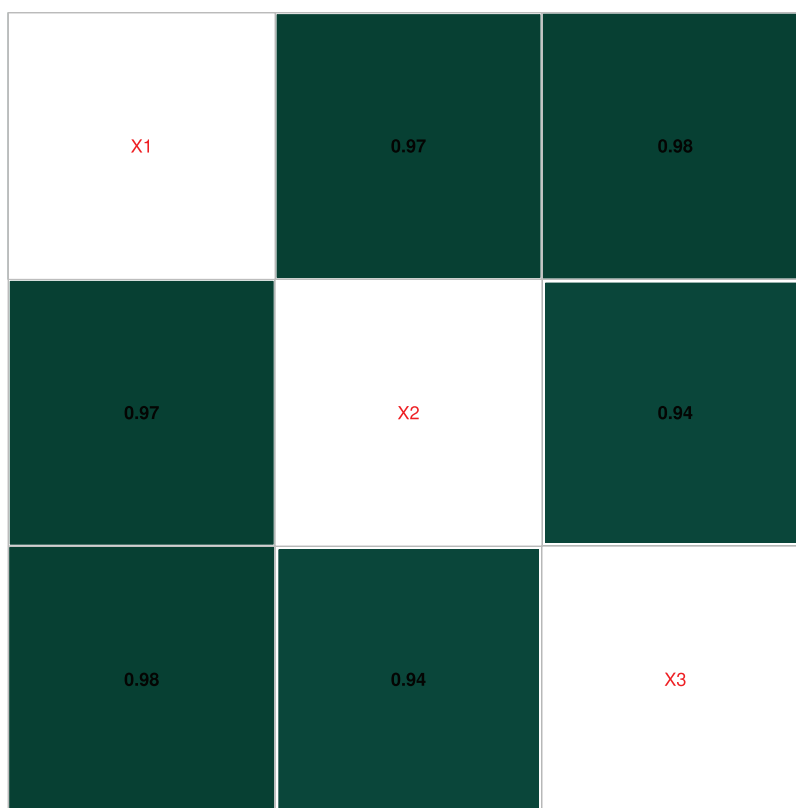


Figure 8. Correlation matrix between explanatory variables in the recreation demand dataset.

Table 11. Estimated coefficients and MSEs of different estimators in the recreation demand dataset.

Estimator	Parameter	β_0	β_1	β_2	β_3	MSE
PMLE	—	1.205252	0.396342	0.125243	−0.804816	1.223553
PRRE	$\hat{k}_1$	1.199927	0.230785	0.151445	−0.660648	0.723359
	$\hat{k}_2$	1.194747	0.139448	0.154931	−0.567259	0.557368
PLRE	$\hat{d}_1$	1.193929	0.128882	0.154453	−0.555295	0.54352
	$\hat{d}_2$	1.194816	0.149823	0.152166	−0.574832	0.568718
PMRTE	$\hat{k}_1, \hat{d}$	1.191583	0.102695	0.152155	−0.524123	0.514341
	$\hat{k}_2, \hat{d}$	1.194747	0.139448	0.154931	−0.567259	0.557368
PARTE	$\hat{k}_1, \hat{d}_1$	1.194374	0.135791	0.154362	−0.562598	0.551995
	$\hat{k}_2, \hat{d}_2$	1.185865	0.084985	0.136734	−0.483919	0.490752
	$\hat{k}_3, \hat{d}_3$	1.203825	0.343343	0.135209	−0.760601	1.035583

inflation of the PMLE variance, which significantly impacts the multicollinearity estimation MSE under conditions of high multicollinearity. The proposed estimator, PARTE, was estimated, its statistical properties were investigated, and its effectiveness was evaluated through theoretical comparisons and extended Monte Carlo simulations using the MSE criterion. The results indicated the impact of sample size, the number of explanatory variables, and the level of multicollinearity on estimating methods. Overall, PARTE demonstrated superiority, exhibiting greater robustness than existing methods in the face

of multicollinearity. Furthermore, the two practical applications provided evidence of PARTE's advantages, as the application results supported the findings of both the simulations and theoretical comparisons. Consequently, the main findings of this study reveal that the PMLE is an unreliable method in the presence of multicollinearity. Finally, the results of both simulation and practical application show that PARTE is the best method for estimating PRM when there is a large amount of multicollinearity. While the proposed PARTE shows performance gains relative to existing biased estimators in the presence of multicollinearity in the PRM, a few limitations should be considered. First, the proposed estimator assumes the correct specification of the Poisson distribution and may suffer from poor performance when the data contain zero-inflation or outliers, in addition to multicollinearity, as is common in count data, which violates the equidispersion assumption of the PRM. Second, the estimator is highly sensitive to the two shrinkage parameters $\hat{k}$ and $\hat{d}$, which control the bias-variance balance in the estimator; these parameters are estimated rather than predetermined, and their selection is crucial to the effectiveness of the estimator, and any inaccuracy in their choice could negate their theoretical benefits. These issues, nevertheless, suggest avenues for future work. Generalization of the framework to other types of regression models, such as quantile regression and zero-inflated models [36,37], would make it more useful in practice. Additionally, cross-validation techniques could be developed as a more empirical approach to choosing the shrinkage parameters, which could enhance the robustness of the estimator across a range of data structures [38]. A robust version of the estimator that simultaneously addresses multicollinearity and the influence of outliers would also be a valuable contribution, offering greater reliability in applied contexts where data rarely conform to ideal conditions [39–41].

Acknowledgments

Princess Nourah bint Abdulrahman University Researchers Supporting Project number (PNURSP2 026R735), Princess Nourah bint Abdulrahman University, Riyadh, Saudi Arabia.

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Data availability statement

The data that supports the findings of this study are available within the article. The data supporting the findings of this study are presented within the article.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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